The semantics and pragmatics of multi-head comparatives

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The version of September 28, 2025

Abstract

A **multi-head comparative** typically contains two comparative expressions and only one *than*-clause: e.g., *Fewer people own more wealth in this country than anywhere else* (Chomsky 1981/1993, von Stechow 1984).

I argue that there is a connection between a **cumulative-reading sentence** like *at most 3% of the population own at least 70% of the wealth* (see Krifka 1999, Brasoveanu 2013) and the multi-head comparative above. This connection is parallel to that between a **measurement sentence** (e.g., *Mary is exactly 6 feet tall*) and a **regular comparative** (e.g., *Lucy is taller than Mary is*): the former conveys the meaning of **measurement**, while the latter expresses a **comparison** between measurements. Thus, a multi-head comparative is the comparative form of a multi-measurement sentence, expressing comparison along multiple dimensions.

I further argue that the multi-head comparative above, along with its corresponding cumulative-reading sentence, addresses a single underlying issue: an issue requiring consideration of measurements or changes along multiple dimensions simultaneously: e.g., an issue like the degree of wealth imbalance involves information about both the population and the amount of wealth they own. Thus, in a multi-head comparative, comparisons along multiple dimensions are not interpreted independently. Rather, they are integrated to make the matrix clause more informative than the *than*-clause in addressing a contextually salient – though potentially implicit – degree QUD (Question under discussion).

Keywords: Comparatives, Multi-head comparatives, Multi-measurement, Cumulative reading, Degree semantics, Degree questions, QUD, Dynamic semantics, Maximality, Informativeness

1 Introduction

A **multi-head comparative** typically contains two comparative expressions and only one *than*-clause, as illustrated in examples (1)–(4), all drawn from von Stechow (1984).

(1) More silly lectures have been given by more boring professors – than I would have expected. (p.42: (136a); originally from Chomsky 1981/1993: p.81, §2.4.4, (6ii))

(2) More dogs ate more rats than cats ate mice. (p.43: (140))

(3) Less land produces more corn than ever before. (p.46: (156))

(4) No airline saves you more money in more ways than Delta. (p.46: (157))

Although multi-head comparatives have been noted for decades in formal syntax and semantics (see, e.g., Chomsky 1981/1993, von Stechow 1984), a thorough investigation is still lacking. On the one hand, von Stechow (1984) suggests that the semantics of multi-head comparatives should be a natural and straightforward extension of our existing understanding of comparatives:

> 'I don't think that these cases represent genuine semantic problems. They can essentially be treated with the methods we already have at our disposal. But it took me quite a while to realize this, because the examples are conceptually rather complicated and are neglected in the literature.'
>
> (von Stechow 1984, pp.41–42)

On the other hand, it has been questioned whether multi-head comparatives are, after all, interpretable and sensible (see, e.g., Hendriks 1994, Hendriks and De Hoop 2001). Even though von Stechow (1984), in the quotation above, denies that they present a genuine challenge, he still characterizes them as 'conceptually rather complicated'.

A corpus search of *the Corpus of Contemporary American English* (https://www.english-corpora.org/coca/, Davies 2008-) reveals that there are indeed naturally occurring sentences that contain multiple comparative expressions:

(5) Henry Ford's assembly line in 1913 increased productivity. Fewer people made more stuff. (2002, NEWS: Atlanta Journal Constitution)

(6) Fewer and fewer people own more and more of the country's land. (2001, MAG: Christian Century)

(7) In the four years of the (Marshall) Plan, the Marshall agency spent $13.5 billion in 16 countries. Fewer people spent more money in that agency than ever before. (1998, TV: Cold War)

(8) As I moved through the years, I saw that I would make more money in more skilled positions. (2012, BLOG: pjmedia.com)

However, the presence of multiple comparative expressions in a sentence does not always guarantee that it is a genuine multi-head comparative, as evidenced by example (9a). Although (9a) contains two comparative expressions and (9b) only one, they are intuitively interpreted in a similar way. Both primarily involve a comparison of school numbers, not a comparison of the length of a school year.

(9) a. More schools choose a longer school year than before.
(adapted from https://www.aps.edu/news/archives/news-from-2021-2022/9-more-aps-schools-choose-a-longer-school-year)
b. More schools choose a long school year than before.

Empirically, what characterizes genuine multi-head comparatives? Theoretically, how to provide a formal analysis? The current paper aims to answer these questions.

In a nutshell, I propose that semantically, a genuine multi-head comparative is the comparative form of a corresponding sentence that expresses **measurements along multiple dimensions simultaneously** (e.g., a cumulative-reading sentence).

Pragmatically, I propose that this **simultaneity** in multi-dimensional comparisons or measurements is not a stipulation, but rather follows naturally from a **single** context-dependent degree QUD (Question-under-discussion, see Roberts 1996/2012). Resolving this degree QUD involves integrating information along multiple dimensions.

For example, the ‘boring-teaching’ sentence in (1) uses both the number of lectures and the number of professors to address, say, a measure of teaching quality; the ‘corn production’ sentence in (3) uses the amount of land and the amount of corn production to address, say, a measure of corn production efficiency (or productivity).

I use the notion of ‘**degree-QUD-based informativeness**’ to analyze both one-dimensional and multi-dimensional measurements or comparisons. For example, having the height ‘$> 6''$’ does not logically entail having the height ‘$= 6''$’. However, in addressing an underlying QUD like ‘how tall X is’, if the tallness degree associated with the measurement ‘$> 6''$’ is reached, it is guaranteed that the tallness degree associated with '= 6″' is reached. Thus, an increase of the height value means an increase of informativeness in addressing the degree QUD of tallness reached.

For both regular and multi-head comparatives, the matrix clause encodes an increase of informativeness based on the *than*-clause, in addressing a contextually salient degree QUD. Regular comparatives in (10) use a change along a single dimension to express an increase of informativeness, while multi-head comparatives in (11) express this increase of informativeness by integrating changes along two dimensions.

(10) **Regular, single-dimensional comparatives: a change (increase or decrease) along one dimension leads to an increase of informativeness**

a. Lucy is taller than Mary is.
-er $\rightsquigarrow$ an increase of informativeness (w.r.t. *tallness Lucy reaches*)

b. Mary is shorter than Lucy is.
-er $\rightsquigarrow$ an increase of informativeness (w.r.t. *shortness Mary reaches*)

c. Mary is less tall than Lucy is.
less $\rightsquigarrow$ an increase of informativeness (w.r.t. *tallness Mary doesn't reach*)

(11) **Multi-head comparatives: changes (increases or decreases) along multiple dimensions together lead to an increase of informativeness**

a. More silly lectures have been given by more boring professors – than I would have expected. (= (1))
more (*lectures*) and *more* (*profs*) combined $\rightsquigarrow$ an increase of informativeness (w.r.t. a salient degree QUD, e.g., a measure of teaching quality)

b. Less land produces more corn than ever before. (= (3))
less (*land*) and *more* (*corn*) combined $\rightsquigarrow$ an increase of informativeness (w.r.t. a salient degree QUD, e.g., a measure of corn productivity)

Thus, multi-head comparatives combine two phenomena already discussed in the existing literature: multi-measurement sentences (e.g., cumulative-reading sentences) and comparatives. This paper further argues that our intuitive interpretation of multi-head comparatives integrates changes along multiple dimensions in addressing a single underlying degree QUD.

Although this contextually salient degree QUD is often implicit, the comparative expressions in a multi-head comparative (e.g., *more* …*more*; *less* …*more*) explicitly convey how changes along two dimensions jointly contribute to higher informativeness in resolving such a QUD. In this sense, these comparative expressions provide explicit cues to what the hidden QUD could be.

In (11a), the use of **two increases** (i.e., *more* ... *more*) evokes a salient degree QUD such that degree information along two dimensions (here the number of lectures and that of professors) contributes to degree-QUD-based informativeness (e.g., a measure of teaching quality) in the same direction. In (11b), the use of **changes in opposite directions** (e.g., *less* ... *more*) evokes a degree QUD such that degree information along two dimensions (here the amount of land and that of corn production) contributes to informativeness (e.g., a measure of productivity) in opposite ways.

The proposed analysis supports von Stechow (1984)'s view that multi-head comparatives do not pose a significant challenge to our existing understanding of comparatives and 'can be treated with the methods we already have at our disposal'.

Conceptually complex phenomena like multi-head comparatives (along with multi-measurement sentences) demonstrate (i) how natural language supports the expression of complex mathematical operations that range from one-dimensional to multi-dimensional comparison and (ii) how the interpretation is guided by an underlying goal and integrates multi-dimensional information into a holistic whole.[1]

Below, based on existing literature (von Stechow 1984, Hendriks 1994, Marques 2005, and Oda 2008a,b), §2 presents detailed empirical observations. §3 connects multi-head comparatives with cumulative-reading sentences. I adopt the insights of Krifka (1999), Brasoveanu (2013), and Zhang (2023) to propose the key ideas of the current analysis. A formal implementation based on dynamic semantics is presented in §4. §5 addresses theoretical implications. §6 concludes.

2 Empirical observations

Empirical observations in this section start with von Stechow (1984)'s discussion of how to interpret typical multi-head comparatives (§2.1). I then examine cases deviating from the typical pattern and consider whether and how they are interpretable (§2.2). The discussion aims to outline the empirical scope of the paper and set the stage for the proposed analysis. Observations concerning the connection between multi-head comparatives and cumulative-reading sentences will be taken up in §3.

[1] Recently, Nouwen (2024)'s work on meiosis and hyperbole as well as Greenberg (2018) and Zhang (2022)'s works on *even* also draw on hidden, contextually salient scales that serve conversational goals.

2.1 Typical multi-head comparatives and their interpretation

For a typical multi-head comparative like (2), von Stechow (1984) describes what it intuitively means and what it does not mean.

As shown in (12), intuitively, the most readily available reading of this sentence invovles **two comparisons**, each along a distinct dimension (see von Stechow 1984, p.43: (141)). Along the dimension of **agent cardinalities**, the dogs that ate rats are compared with the cats that ate mice (see (12a)). Along the dimension of **theme cardinalities**, the rats eaten by dogs are compared with the mice eaten by cats (see (12b)).

(12) **von Stechow (1984): what a multi-head comparative intuitively means:**

$\underbrace{\text{More dogs}}_{\text{agent}}$ ate $\underbrace{\text{more rats}}_{\text{theme}}$ than $\underbrace{\text{cats}}_{\text{agent}}$ ate $\underbrace{\text{mice}}_{\text{theme}}$. (= (2))

a. The comparison of **agent cardinalities**: |dogs| vs. |cats|
The number of the dogs that ate rats > the number of the cats that ate mice

b. The comparison of **theme cardinalities**: |rats| vs. |mice|
The number of the rats eaten by dogs > the number of the mice eaten by cats

For this reading described in (12), multiple comparisons take place along distinct dimensions, and these dimensions (as well as the comparisons along each) are mutually independent (see (12a) vs. (12b)).

Furthermore, von Stechow (1984) notes that there is a **mutual restriction** between the agent and the theme, at the matrix-clause and the *than*-clause level respectively. In interpreting this sentence in (12), we do not compare the total number of dogs and cats in the context. Rather, we count only the dogs that ate rats and the cats that ate mice, and then compare the cardinalities of these restricted dog-sum and cat-sum. Similarly, it is the rats eaten by dogs and the mice eaten by cats (the restricted rat-sum and mouse-sum) whose cardinalities undergo comparison. This mutual restriction is also the hallmark of cumulative reading and will be discussed in §3.

According to von Stechow (1984), the interpretation in (13) does not accurately capture our intuitive understanding of the sentence (see von Stechow 1984, p.43: (142)). In contrast to (12), which expresses two mutually independent comparisons along two dimensions, (13) only expresses **one comparison**: between the cardinality of all dog-rat pairs involved in eating-events and that of all cat-mouse pairs involved in such events.

(13) More dogs ate more rats than cats ate mice. (= (2))

von Stechow (1984): what this multi-head comparative does not mean:
The number of pairs $\langle x, y\rangle$ > the number of pairs $\langle x', y'\rangle$
(where $\textsc{dog}(x), \textsc{rat}(y), \textsc{cat}(x'), \textsc{mouse}(y'), \textsc{eat}(x, y), \textsc{eat}(x', y')$)

This event-counting-based comparison in (13) is too weak. Under the scenario in (14), where *more dogs ate more rats than cats ate mice* is intuitively judged false, the reading described in (13) nevertheless comes out as true, contradicting our intuitive judgment.

(14) Scenario: Dog_1 ate Rat_1, Rat_2, and Rat_3; Cat_1 ate Mouse_1, and Cat_2 ate Mouse_2.
Thus there are 3 'dog-eating-rat' events and 2 'cat-eating-mouse' events.

2.2 Deviations from typical multi-head comparatives

Hendriks (1994) (see also Hendriks and De Hoop 2001) argues that von Stechow (1984)'s account over-generates: Not all sentences with multiple comparative expressions are interpretable as expressing independent comparisons along distinct dimensions, as shown in (12). Some such sentences are in fact intuitively nonsensical.

Here I examine 6 types of deviations from (12), investigating whether and to what extent they are interpretable and what meaning they have.

2.2.1 Comparative expressions with opposite directions of inequalities

While von Stechow (1984)'s analysis in (12) is expected to extend to cases like (15), Hendriks (1994) argues that (15) is intuitively nonsensical.

(15) *$\underbrace{\text{Fewer dogs}}_{\text{agent}}$ ate $\underbrace{\text{more rats}}_{\text{theme}}$ than $\underbrace{\text{cats}}_{\text{agent}}$ ate $\underbrace{\text{mice}}_{\text{theme}}$. (Hendriks 1994: (6))

The analysis à la von Stechow (1984): two comparisons (Hendriks 1994: (6′))

a. Along the dimension of agent cardinalities:
The number of the dogs that ate rats < the number of the cats that ate mice

b. Along the dimension of theme cardinalities:
the number of the rats eaten by dogs > the number of the mice eaten by cats

Is the degradedness of (15) due the opposite directions of the two inequalities (see (15a) vs. (15b))? The existence of data like (3) (repeated as (16)) suggests that the answer is *no*. (16) involves two comparisons in opposite directions, but (16) is an intuitively natural and acceptable multi-head comparative.

(16) Less land produces more corn than ever before. (= (3))

Why, then, does our intuition about (15) contradict von Stechow (1984)'s prediction? Hendriks (1994) argues that the two comparisons in a multi-head comparative cannot be mutually independent, and proposes the analysis in (17) as the intended meaning of the degraded sentence (15).

(17) **Hendriks (1994)'s analysis of (15)** (see Hendriks 1994: (8)):
the number of dogs x such that '|rats eaten by x| > |mice eaten by y|' <
the number of cats y such that '|rats eaten by x| > |mice eaten by y|'

In (17), one comparison is embedded within another. The inner comparison '|rats eaten by x| > |mice eaten by y|' restricts the items x and y, which are further compared along the dimension of cardinality. Hendriks (1994) claims that this restriction '|rats eaten by x| > |mice eaten by y|' makes x and y mutually dependent and creates infinite regress: the definition of x relies on y, and vice versa. Thus, this mutual dependency cannot be derived compositionally, and the meaning of (15) is judged nonsensical.[2]

Hendriks (1994)'s analysis of the degradedness of (15) (as shown in (17)) is questionable. Many multi-head comparatives are intuitively natural and interpretable. Clearly, their semantic derivation does not suffer from the problem of infinite regress. Thus multi-head comparatives can involve comparisons in opposite directions. The degradedness of examples like (15) needs a different explanation.

Meier (2001) offers a relevant observation: 'The fact that (15) is somehow odd is just a reflex of the fact that it is hard to imagine a situation in which the comparison construction might be relevant (p.356).' Indeed, (18) illustrates that a sentence similar to (15) sounds natural when used in an appropriate context.[3]

(18) Context: A zoo keeper is discussing the cost of feeding animals. A lion eats between 5 and 7 kg of meat per day, while a chimpanzee typically eats fruit, with a daily intake of 1 to 4 kg. Feeding and keeping lions is more costly.
Fewer lions eat more meat than chimpanzees eat fruit. (cf. (15))

[2]In §3 and §4, I will show that this kind of mutual restriction does not challenge semantic compositionality. Works like Brasoveanu (2013) have provided a solution (see also Charlow 2017). Beyond the literature on cumulative reading, Bumford (2017) investigates the mutual definition of uniqueness in Haddock descriptions (e.g., *the rabbit in the hat*, see Haddock 1987) and provides a related analysis.

[3]I thank Takeo Kurafuji for discussing the example (15) with me and providing this scenario in (18).

Under the context in (18), our interpretation of this multi-head comparative can be naturally explained by an analysis à la von Stechow (1984). (18) involves two comparisons, one along the dimension of animal cardinalities and the other along the amount of food consumption. The opposite directions of the inequalities indicate that feeding lions ranks lower than feeding chimpanzees on a scale of cost-effetiveness.

2.2.2 Multi-head comparatives with proportional or distributive reading

As discussed above, von Stechow (1984)'s analysis shows that multi-head comparatives involve two comparisons along two distinct dimensions: one measuring and comparing agents, and the other measuring and comparing themes. These measurements typically count the **cardinality or total amount** of the relevant agents and themes. However, multi-head comparatives are not limited to such measurements, as evidence by the existence of **proportional** and **distributive-reading** multi-head comparatives.

(19) and (20) are naturally occurring examples of multi-head comparatives with a proportional reading. Based on our world knowledge, the most natural interpretation involves **percentages** rather than cardinalities. In this way, multi-head comparatives, like regular comparatives, allow for flexibility with regard to measurement units (see (21)).

(19) The trend is indisputable: Fewer people own more of the overall wealth, and fewer companies own more market share.
(https://www.deseret.com/opinion/2020/9/14/21436415/guest-opinion-america-capitalism-strengths-dark-side-too-far-inequality-divisiveness-wealth-gap)

(20) Fewer people own more of the land in Brazil than anywhere else in the world.
(https://glenmorangie.newint.org/features/2003/01/05/cutting)

(21) More people are hired here than elsewhere. **cardinalities ✓; percentages ✓**

The flexibility of measurement units is also reflected in distributive-reading multi-head comparatives. Marques (2005) and Oda (2008a,b) have noted this phenomenon in Portuguese and Japanese (see (22) and (23)). In these examples, the measurement and comparison along one dimension is based on cardinalities (i.e., the number of predators in (22) and the number of companies in (23)), while that along the other dimension is based on per-unit values (i.e., the number of preys per predator in (22) and the amount of corporate tax per company in (23)). The distributive meaning is overtly expressed by the use of *sorezore* 'each' in (22), but covertly inferred from our world knowledge in interpreting (23).

(22) **Distributive-reading multi-head comparative in Japanese**

San-biki-no	neko-ga	**sorezore**	yon-hiki-no	hatukanezumi-o	tabeta	yorimo	
3-CL-GEN	cat-NOM	each	4-CL-GEN	mouse-ACC	ate	THAN	
(motto)	takusanno	inu-ga	**sorezore**	(motto)	takusanno	dobunezumi-o	tabeta.
(more)	many	dog-NOM	each	(more)	many	rat-ACC	ate

Lit. 'More dogs ate more rats each than three cats ate four mice each.'
⇝ There are 3 cats and each of them ate 4 mice. There are more than 3 dogs and each of them ate more than 4 rats. (Oda 2008b: (62) and (63))

(23) **Distributive-reading multi-head comparative in Portuguese**

Eles	querem	que	mais	empresas	paguem	menos	IRC
they	want	that	more	companies	pay	less	corporate-tax

'They want more enterprises to pay less corporate tax.' (Marques 2005: (65))

Naturally occurring distributive-reading multi-head comparatives also exist in English, as illustrated in (24). Based on world knowledge, *more of their time* in (24) should be measured and compared per person, resulting in a distributive reading.

(24) The world would be a more positive place to be if more people spent more of their time focusing on what they're thankful for.
(https://schizanthusnerd.com/2018/04/01/everyday-gratitude/)

2.2.3 Comparative expressions beyond plural/mass nouns

von Stechow (1984) suggests that comparative expressions in a multi-head comparative are restricted to plural or mass nouns (e.g., *more dogs* and *more rats* in (2), *less land* in (3)). A sentence like (25) is ungrammatical, because the comparative expressions *greater* and *better* are not plural or mass nouns. Hendriks (1994) presents another ungrammatical example in (26), where the comparative expression *higher* is not a plural or mass noun.

(25) *A greater man would be a better man than Otto. (von Stechow 1984: p.46, (158))

(26) *$\underbrace{\text{More doors}}_{\text{items}}$ are $\underbrace{\text{higher}}_{\text{linear size}}$ than $\underbrace{\text{windows}}_{\text{items}}$ are $\underbrace{\text{wide}}_{\text{linear size}}$. (Hendriks 1994: (5))

According to von Stechow (1984), in a multi-head comparative, each comparative expression needs to be associated with an overt or silently reconstructed *than*-clause. For a sentence like (25), von Stechow (1984) claims that the reconstruction gives rise to two possibilities in interpreting the *than*-phrase (see (27a) and (27b)): 'the low acceptability of (25) shows that our grammar doesn't work that way. It seems, then, that the restrictions for the reconstruction of a *than*-phrase are rather syntactic than semantic.'

(27) *A greater man (**than Otto**) would be a better man **than Otto**.

a. than Otto is a great man

b. than Otto is a good man

von Stechow (1984)'s reconstruction-based explanation in (27) is challenged by examples like (28), which contains two comparative expressions that are not plural or mass nouns. The grammaticality of (28) shows that there is in principle no problem in associating multiple comparative expressions with the same *than*-clause/phrase. The interpretation of (28) involves two comparisons, along the dimensions of greatness and goodness, and the elided gradable adjectives in the *than*-clause are recovered just as in a regular comparative that involves only a single comparative expression.

(28) He is a greater and better man than I am. $\rightsquigarrow$ than I am ~~a d-great and d'-good man~~

There are more examples challenging von Stechow (1984)'s 'plural/mass nouns' restriction, as shown in (29)–(31) (e.g., *faster* in (29), *better* in (30) and (31)). Thus, the unacceptability of (25) and (26) requires a different explanation (see §§2.2.4 and 2.2.6).

(29) Nowadays, more goods are carried faster (than before).
(Hendriks and De Hoop 2001: p.10, (13))

(30) Aldi's pork selection was a little more varied with better value than what I saw here. (https://www.insider.com/aldi-vs-lidl-review-differences-which-better-photos-2021-5)

(31) Hydrogen-powered cars also provide slightly more driving range with better energy density than batteries and fuel. (https://www.whichcar.com.au/car-advice/hydrogen-cars-v-electric-cars-australia)

2.2.4 The lack of parallelism between the matrix and the *than*-clause

von Stechow (1984) notes that (32) is intuitively weird as a multi-head comparative. He argues that the analysis of a multi-head comparative requires silent reconstruction of a *than*-clause, but (32) is different from (1) (repeated as (33)): reconstructing the *than*-clause for the first comparative expression in (32) is impossible (see (34)).

(32) *More silly lectures have been given by more boring professors than I met yesterday. (Chomsky 1981/1993: p.81, §2.4.4, (6iii))

(33) More silly lectures have been given by more boring professors – than I would have expected. (=(1))
⇝ More silly lectures (**than I would have expected that silly lectures would be given by boring professors**) have been given by more boring professors **than I would have expected (that silly lectures would be given by boring professors)**.

(34) *More silly lectures (**than I met yesterday**) have been given by more boring professors **than I met yesterday**. Reconstructing the *than*-clause for (32)

von Stechow (1984)'s account for the ungrammaticality of (32) is based on the ungrammaticality of (34): *meet* is compatible with nouns like *professors*, but not nouns like *lectures*. If *meet* in (34) is replaced by *evaluate*, a verb compatible with both *lectures* and *professors*, then a sentence like (35), which is parallel to (34), is acceptable.

(35) Context: As chair of the academic committee, I evaluate courses and professors annually. To my surprise, teaching quality has declined significantly this year — both in terms of the number of silly lectures and the number of boring professors.
More silly lectures than I evaluated last year have been given by more boring professors than I evaluated last year.

If a multi-head comparative indeed involves reconstructing a *than*-clause for the first comparative expression, we would predict that (36) should be as acceptable as (35). But this prediction is not borne out. Intuitively, (36) is as weird as (32).

(36) *More silly lectures have been given by more boring professors than I evaluated last year.

Thus the contrast between the ungrammatical multi-head comparative (36) and the acceptable sentence in (35) argues against von Stechow (1984)'s reconstruction-based analysis. The interpretation of multiple comparisons in a multi-head comparative should not be based on a distinct, independent *than*-clause for each comparative expression.

Rather, the comparison standards needed in the interpretation of multiple comparisons should all be recoverable from the same *than*-clause. In other words, parallelism is needed between the matrix and the *than*-clause in providing information on multiple measurements for multi-dimensional comparison.[4]

As illustrated in (37), for a regular, one-dimensional comparative (see (37a)), both the matrix and the *than*-clause provide single-dimensional measurements. Similarly, for a multi-head comparative that involves multi-dimensional comparisons (see (37b)), both the matrix and the *than*-clause provide parallel multi-dimensional measurements.

(37) **Parallelism between what undergoes comparisons**:

a. $\underbrace{\text{Lucy is taller } \underbrace{\text{than Mary is } \cancel{\text{tall}}}_{\text{comparison standard: height}}}_{\text{comparison target: height}}$ **One-dimensional**

b. $\underbrace{\text{More dogs ate more rats } \underbrace{\text{than cats ate mice}}_{\text{standards: agent and theme cardinalities}}}_{\text{targets: agent and theme cardinalities}}$ **Two-dimensional**

Thus sentences like (32) and (36) are weird because their *than*-clause (i.e., *than I met ~~d-many professors~~ yesterday* in (32), *than I evaluated ~~d-many professors~~ last year* in (36)) does not provide multi-dimensional measurements, failing to support the parallelism between the matrix and the *than*-clause to conduct multi-dimensional comparison.

In §2.2.3, (25) (**a greater man would be a better man than Otto*) is unacceptable for the same reason: there is no parallelism between the *than*-expression, *than Otto*, and the matrix clause, *a greater man would be a better man*. In contrast, in interpreting (28) (i.e., *he is a greater and better man than I am ~~a d-great and d′-good man~~*), parallelism is available.

[4]*More people have been to Russia than I have* is a famous illusion that violates this parallelism requirement (see Phillips et al. 2011). More broadly, natural language contains other comparison-related illusions: e.g., *no head injury is too trivial to be ignored* (see Hohaus and Bade 2022); *the greatest happiness for the greatest number* (see Morzycki 2018 on multi-head superlatives). In addition, §2.2.6 discusses what I call 'pseudo-multi-head comparatives', and cases like (28) are not central to the current investigation.

Despite various complications, this paper argues, based on a detailed examination of data, that genuine multi-head comparatives are not illusions. Instead, they allow for and require a compositional analysis that derives their most salient reading. The empirical generalizations of this core set of multi-head comparatives are summarized in §2.3. I thank an anonymous reviewer for raising the issue of grammatical illusions.

2.2.5 Multi-head comparatives without an overt *than*-clause

Sometimes, multi-head comparatives are like regular comparatives in containing no overt *than*-clause (see (38)). Thus comparatives never syntactically require a *than*-clause.

(38) a. (Mary is exactly 6 feet tall.) Lucy is taller. **Regular comparative**
b. Newer generations of microchips contain more electronic switches on a smaller surface. **Multi-head comparative** (Hendriks 1994: (14))

Evidently, in a comparison with the target (e.g., Lucy's height in (38a)), the standard (e.g., Mary's height in (38a)) can be from either (i) an overt *than*-clause/phrase in the same sentence or (ii) a discourse-salient item in the context.

Hendriks (1994) suggests that **discourse comparatives** (i.e., multi-head comparatives without an overt *than*-clause) are more interpretable than multi-head comparatives with an overt *than*-clause. She further claims that 'comparatives may contain at most one instance of *sentence-internal* comparison (Hendriks 1994: (17))'. Here *sentence-internal* comparison means comparison based on an overt *than*-clause.

Examples like (12) (i.e., *more dogs ate more rats than cats ate mice*) clearly contradict this view. In (12) the *than*-clause (*than cats ate mice*) supports two sentence-internal comparisons: along the dimensions of agent and theme cardinality.

It is likely that, in the absence of an overt *than*-clause, parallelism between targets and standards along multi-dimensional comparison can be more easily accommodated. By contrast, the presence of an overt *than*-clause sometimes blocks this parallelism (e.g., *more silly lectures have been given by more professors* (**than I met*)).

The role of comparative expressions (especially the meaning of *-er*/*more*) and how to interpret comparatives with or without an overt *than*-clause will be addressed in §4.2.

2.2.6 Pseudo-multi-head comparatives that conduct only one comparison

According to Hendriks (1994), a multi-head comparative with an overt *than*-clause involves only one comparison, due to (i) the issue of mutual dependency and infinite regress (see §2.2.1) and (ii) the constraint of 'at most one instance of sentence-internal comparison (Hendriks 1994: (17))' (see §2.2.5), as illustrated in (39).

(39) John made more people prettier than I thought he would. (Hendriks 1994: (12))
$\rightsquigarrow$ John made more people prettier than I thought he would ~~make m-many people prettier~~. (Hendriks 1994: (16))

In (39), the *than*-clause measures how many people I thought he would make prettier. The matrix clause measures how many people John actually made prettier. So (39) expresses only one comparison: between human cardinalities.

Similarly, in our most natural interpretation of sentence (9a) (repeated as (40)), both the matrix and the *than*-clause provide information on how many schools choose a longer school year now / before. Only one comparison is made: between school numbers.

(40) <u>More</u> schools choose a longer school year <u>than</u> before. (= (9a))
⇝ More schools choose a longer school year than ~~*m*-many schools chose a longer school year~~ before

Thus, (39) and (40) should not be considered genuine multi-head comparatives. Containing multiple comparative expressions is a necessary but not sufficient condition for a genuine multi-head comparative. Genuine multi-head comparatives involve simultaneous comparisons along different dimensions, and their interpretation requires integrating information along multiple dimensions. For instance, increases along both the number of silly lectures and boring professors contribute jointly to assessing an issue like teaching quality; an increase in corn production combined with a decrease in land use reflects improved productivity.

This explains why (26) (repeated as (41)) is infelicitous (see §2.2.3). It is difficult to accommodate a natural context in which interlocutors care about both the number and the size of doors / windows for their conversational goal (see also Meier 2001).

(41) *More doors are higher than windows are wide. (= (26))

2.3 Interim summary

To sum up, this section has shown empirically what characterizes a genuine multi-head comparative. A genuine multi-head comparative involves simultaneous comparisons along multiple comparisons (§2.1). These multi-dimensional comparisons can have the same or opposite directions (§2.2.1), allow flexibility in measurement units (§2.2.2), and are not restricted to comparisons of quantity (cf. the 'plural/mass nouns' restriction in §2.2.3). Comparison standards can be overly expressed or inferred from context (§2.2.5).

Crucially, a genuine multi-head comparative requires parallelism between comparison standards (from the *than*-clause or context) and comparison targets (from

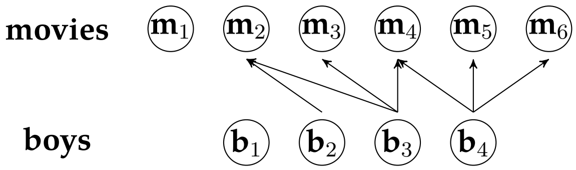

Figure 1: The genuine **cumulative** reading of (42) is **true** in this context.

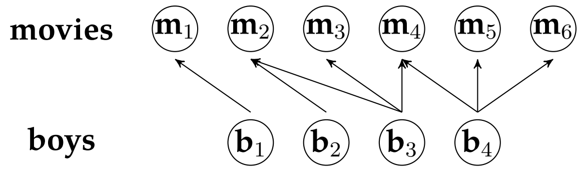

Figure 2: The genuine **cumulative** reading of (42) is **false** in this context.

the matrix clause) in providing multi-dimensional measurements (§2.2.4). Moreover, these multi-dimensional measurements need to allow for holistic integration, contributing to the resolution of a single, contextually salient issue (§2.2.6).

This kind of information integration across multiple dimensions closely parallels what we observe in cumulative-reading sentences. In what follows, I develop an analysis of multi-head comparatives informed by existing work on cumulative-reading sentences.

3 Inspiration from cumulative-reading sentences

I first present Brasoveanu (2013)'s analysis of cumulative-reading sentences, focusing on the simultaneity of multi-dimensional measurements (§3.1). Then I follow Krifka (1999) and Zhang (2023)'s discussion to show the interplay between multi-dimensional measurements: they jointly address a single underlying issue. Based on these insights, an informal explanation of multi-head comparatives is presented in §3.3.

3.1 Brasoveanu (2013)'s analysis of cumulative-reading sentences

A sentence like (42) contains two modified numerals (see the underlined parts) and has both a **distributive** and a **cumulative** reading (see (42a) and (42b)).

(42) <u>Exactly three boys</u> saw <u>exactly five movies</u>. (see e.g., Brasoveanu 2013)

a. **Distributive**: There are exactly 3 boys such that each of them saw exactly 5 movies.

b. **Cumulative**: Some boys saw some movies. The total number of boys who saw any movies is 3, and the total number of movies seen by any boys is 5.

Krifka (1999) and Brasoveanu (2013) emphasize that the derivation of the genuine cumulative reading (42b) is necessarily built on a **mutual restriction** between the two modified numerals, so that the two modified numerals are interpreted simultaneously and no scope-taking is involved.

As shown in (43), if one modified numeral (here *exactly three boys*) takes scope over the other (here *exactly five movies*), then only the latter is involved in restricting and defining the former, but not vice versa. The reading derived in (43) (dubbed the pseudo-cumulative reading) is true under the scenario in Fig. 2.

(43) **Unattested pseudo-cumulative reading** of (42): The maximal boy-sum such that 'they saw in total five movies between them' has a cardinality of 3.
$\rightsquigarrow$ True under the scenario in Fig. 2: there are two boy-sum witnesses, $\mathbf{b}_2 \oplus \mathbf{b}_3 \oplus \mathbf{b}_4$ and $\mathbf{b}_1 \oplus \mathbf{b}_2 \oplus \mathbf{b}_4$. There is no larger boy-sum satisfying the restriction that 'they saw in total five movies between them'.

Intuitively, sentence (42) is false under the scenario in Fig. 2, indicating that the pseudo-cumulative reading in (43) is too weak and unattested. Thus the genuine cumulative reading of (42) must be stronger and involve mutual restriction.

The genuine cumulative reading (see (42b)) is true under the scenario in Fig. 1. For this reading, *exactly three boys* denotes and counts the totality of **boys who saw any movies**, which is 3 (cf. the cardinality of all boys in the context is 4), and *exactly five movies* denotes and counts the totality of **movies seen by any boys**, which is 5 (cf. the cardinality of all movies in the context is 6).

Brasoveanu (2013) adopts **dynamic semantics** to implement these ideas and develops a formal analysis for the cumulative reading of (42). Essentially, modified numerals make semantic contributions in several layers, as sketched in (44):

(44) Exactly three boys saw exactly five movies. **Cumulative reading** of (42)

$$\underbrace{\sigma x \sigma y[\textsc{boy}(x) \wedge \textsc{movie}(y) \wedge \textsc{saw}(x,y)]}_{\text{the mereologically maximal } x \text{ and } y} \wedge \underbrace{|y| = 5 \wedge |x| = 3}_{\textbf{post-suppositional} \text{ cardinality tests on selected drefs}}$$

First, each modified numeral introduces a potentially plural discourse referent (dref). Then restrictions are added onto drefs: here $\textsc{boy}(x)$, $\textsc{movie}(y)$, and $\textsc{saw}(x,y)$.[5]

Second, at the sentence level, after all the drefs are introduced and restrictions are added, modified numerals contribute **mereology-based maximality operators** – σ (similar to a Link-style sum operator). These maximality operators are applied at the sentence level simultaneously, picking out the maximal drefs satisfying the restrictions

[5] Cumulative closure is assumed for lexical relations when needed.

BOY(x), MOVIE(y), and SAW(x, y).

Finally, the modified numerals contribute **post-suppositional cardinality tests**, checking whether the cardinality of the **selected** maximal boy-sum (i.e., those who saw movies) is 3 and whether the cardinality of the relativized maximal movie-sum is 5.

Evidently, there is a great similarity between cumulative-reading sentences and multi-head comparatives (see von Stechow 1984's analysis of *More dogs ate more rats than cats ate mice* in (12) in §2.1). Our intuitive interpretation of both involves mutual restriction and relativized maximization.

Mutual restriction paves the ground for identifying the relativized maximal plural individuals. A cumulative-reading / multi-measurement sentence then **measures the cardinality of each relativized maximal plural individual** (see (45a)), whereas a multi-head comparative **compares these cardinalities** (see (45b)).

(45) **Measurement and comparison along two dimensions: #agent and #theme**

a. Exactly three cats ate exactly five mice.
(i) selecting the relativized maximal cat-sum and mouse-sum;
(ii) checking their measurements, i.e., **checking cardinalities**

b. More dogs ate more rats than cats ate mice
(i) selecting the relativized maximal dog-, rat-, cat-, and mouse-sums;
(ii) comparing their measurements, i.e., **checking** whether the cardinality of the dog-sum is **larger** than that of the dog-sum, and the cardinality of the rat-sum is **larger** than that of the mouse-sum, respectively

The connection between multi-measurement sentence (45a) and multi-head comparative (45b) is thus parallel to that between measurement sentence (46a) and comparative (46b). For regular measurement sentences and comparatives, measurement and comparison are conducted along a single dimension (e.g., the cardinality of the theme in (46)). For multi-measurement sentences and multi-head comparatives, measurement and comparison are conducted simultaneously along multiple dimensions (e.g., the cardinalities of the agent and the theme in (45)). Multi-head comparatives like (45b) are the comparative form of multi-measurement sentences.

(46) **Measurement and comparison along one dimension: the cardinality of theme**

a. Mary bought 6 books.

b. Lucy bought more books than Mary did ~~buy m-many books~~

3.2 Interplay between multi-dimensional measurements

Krifka (1999) discusses cumulative-reading sentences like (47), arguing that the **simultaneous mereology-based maximization** strategy (used in analyzing sentences like (42) – *exactly 3 boys saw exactly 5 movies*) does not apply to (47):

(47) In Guatemala, (at most) 3% of the population own (at least) 70% of the land.

Krifka (1999) points out that

> 'this strategy would lead us to select the alternative *In Guatemala, 100 percent of the population own 100 percent of the land*, which clearly is not the most informative one among the alternatives – as a matter of fact, it is pretty uninformative.'
>
> 'What is peculiar with sentences like (47) is that they want to give information about the bias of a statistical distribution. One conventionalized way of expressing particularly biased distributions is to select a small set among one dimension that is related to a large set of the other dimension.'
>
> (Krifka 1999: §3.1)

Krifka (1999)'s discussion suggests that informativeness is not determined by numeric values alone. For (47), increasing the number values (e.g., from '≤ 3%' and '≥ 70%' to '100%' for both) does not result in higher informativeness, because here informativeness should be about a 'biased distribution'.

Fintel et al. (2014) makes related observations, distinguishing between **upward-** and **downward-monotonic** degree properties. For an upward-monotonic property, as in (48a), increasing the number increases informativeness. For a downward-monotonic property, as in (48b), informativeness increases when the number decreases.

(48) a. **Degree properties that are upward-monotonic** (Fintel et al. 2014):
$\lambda n.$ Miranda has n kids
e.g., Miranda has 4 kids $\underline{>_{\text{INFO}} \text{ (i.e., } \textbf{entails})}$ Miranda has 3 kids

b. **Degree properties that are downward-monotonic** (Fintel et al. 2014):
$\lambda d.d$-much walnuts is sufficient to make a pan of baklava
e.g., 300 g of walnuts are sufficient to make a pan of baklava
$\underline{>_{\text{INFO}} \text{ (i.e., } \textbf{entails})}$ 500 g of walnuts are sufficient to make a plan of baklava

Zhang (2023) further argues that there should be a distinction between **logical entailment** and **informativeness**.[6] As shown in (49), for a degree property like (48a), increasing or decreasing the numbers does not always guarantee logical entailment.

(49) a. Miranda has exactly 4 kids $\not\models$ Miranda has exactly 3 kids
b. Miranda has exactly 3 kids $\not\models$ Miranda has exactly 4 kids

Nevertheless *Miranda has exactly 4 kids* expresses a higher degree in a context that addresses, e.g., the burden of raising kids (see (50)). Here informativeness is not based on logical entailment tied to a **context-independent numerical scale**, but rather on entailment relative to a **context-dependent scale**, e.g., burden level. Having exactly four kids is guaranteed to reach a higher degree of burden than having exactly three.

(50) Context: The more kids one has, the heavier the burden is.
Miranda has exactly 4 kids $>_{\text{INFO}}$ Miranda has exactly 3 kids
$\rightsquigarrow$ '$|\text{kids}| = 4$' corresponds to a degree of burden higher than '$|\text{kids}| = 3$'

This kind of informativeness does not depend on logical entailment (as in (48a)), but on a contextually salient degree QUD that reflects interlocutors' underlying concern.

When people care about, e.g., the burden of raising kids, increasing the number of kids (from 3 to 4) leads to a heavier burden, i.e., higher informativeness. But if a fixed stipend is granted to anyone with at least three children, then *Miranda has exactly **three** kids* is as informative as *Miranda has exactly **four** kids* with respect to stipend eligibility.

Similarly, for cumulative-reading or multi-measurement sentences, informativeness is based on an underlying, context-dependent issue. As Krifka (1999) points out, the sentence *At most 3% of the population own at least 70% of the land* 'gives information about the bias of a statistical distribution'. It is this kind of underlying issue that determines how changes of numerical values lead to higher or lower informativeness.

As sketched out in (51), different multi-measurement sentences assume different kinds of underlying, contextually salient issues, so that numerical measurements do not contribute to informativeness in the same way. Generally, context leads to two patterns of interplays: (i) measurements along both dimensions contribute to informativeness in the same direction (see (51a)); (ii) they contribute in opposite directions (see (51b)). In either case, this interplay addresses a single context-dependent issue.

[6] Fintel et al. (2014) also mentioned in their footnote 4 that it is likely that some notion of informativity other than logical entailment is needed.

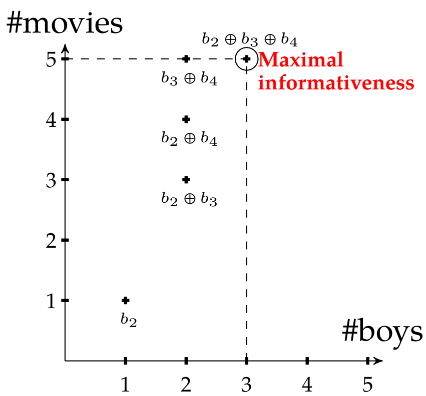

Figure 3: (42): *Exactly three boys saw exactly five movies*.
The cardinalities of boy-sums and the movies they saw are plotted.
Maximal informativeness corresponds to mereological maximality along both dimensions.

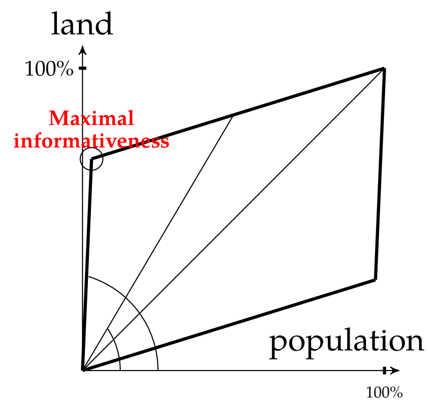

Figure 4: (47): *At most 3% of the population own at least 70% of the land*.
The percentages of land-owning populations and the total land they own are plotted as a parallelogram-shaped area.
Maximal informativeness corresponds to the largest ratio.

(51) a. Exactly three boys saw exactly five movies. (= (42))
⇝ Maximal informativeness corresponds to maximal values along both dimensions (see Fig. 3)[7]

b. At most 3% of the population own at least 70% of the land. (= (47))
⇝ Maximal informativeness corresponds to the maximal ratio (see Fig. 4)

Oftentimes, this underlying issue is not overtly expressed or specified, but inferred from context and our world knowledge. That said, the monotonicity of the modified numerals in a multi-measurement sentence does offer overt clues about how maximal informativeness is calculated and what kind of issue is being assumed.

When two numerals contribute to informativeness in the same direction (see (51a)), they usually share the same monotonicity. In contrast, for a ratio-based interpretation (see (51b)), the two numerals contribute to informativeness in opposite ways, and thus often exhibit opposite monotonicity. In (51b), *at most 3%* is downward-entailing, while *at least 70%* is upward-entailing.

[7]In Brasoveanu (2013)'s analysis (see §3.1), the simultaneous application of mereology-based maximality operators actually reflects this implicit assumption of a relevant contextually salient issue.

3.3 From multi-measurement sentences to multi-head comparatives

A multi-head comparative is the comparative of a multi-measurement sentence, integrating multi-dimensional information to address a single context-dependent issue.

For a regular comparative (see (52)), its matrix and *than*-clause offer parallel measurements along a single dimension to address a degree QUD (here *how many kids one has*) with maximal informativeness (see (52a)). Then inequality is checked, just like cardinalities are checked for a cumulative-reading sentence (see (52b)).

(52) Miranda has more kids (than Betty ~~has n-many kids~~)

a. maximal informativeness at the matrix and *than*-clause level
$\leadsto$ select (i) the maximal sum of kids that Miranda has and (ii) the maximal sum of kids that Betty has, and count their cardinalities

b. check whether inequality, '>', between these two cardinalities holds true

For multi-head comparatives (see (53) and (54), along with Fig. 5 and 6), their matrix and *than*-clause offer parallel multi-dimensional measurements to address a single contextually salient issue ((53a) and (54a)). Inequalities along each dimension are then checked, just like cardinality tests in cumulative-reading sentences ((53b) and (54b)).

(53) More dogs ate more rats than ~~m-many~~ cats ate ~~n-many~~ mice. (see Fig. 5)

a. maximal informativeness at the matrix and *than*-clause level
$\leadsto$ select the cases that yield maximal informativeness (the maximal rat-eating dog-sum, etc.) and measure them

b. check whether inequalities, '>' and '>', hold true along the two dimensions

(54) Fewer people own more wealth here than ~~m% of the population own n% of the wealth~~ elsewhere. (see Fig. 6)

a. maximal informativeness at the matrix and *than*-clause level
$\leadsto$ select the cases that yield maximal informativeness (i.e., where the ratio between population and wealth owned is maximal) and measure them

b. check whether inequalities, '<' and '>', hold true along the two dimensions

Just as the monotonicity of modified numerals in cumulative-reading sentences helps reveal the relevant QUD (see §3.2), in multi-head comparatives, the polarity of comparative expressions (e.g., *more*, *fewer*) offers overt clues about the contextual QUD.

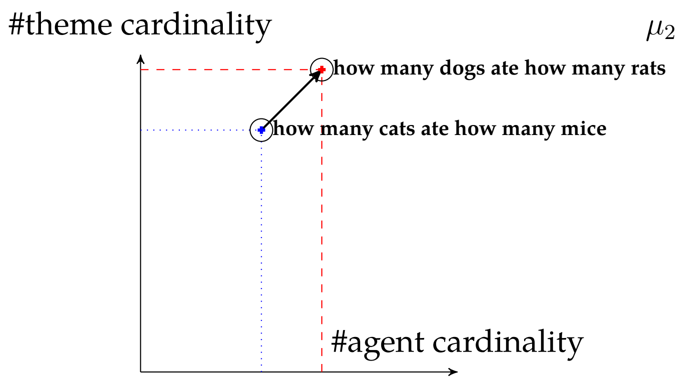

Figure 5: *More dogs ate more rats than cats ate mice* (see (53)).
The two highlighted dots represent the cases of maximal informativeness at the matrix and *than*-clause level: maximal informativeness involves maximal measurement along each dimension.
Then the inequalities along each dimension are checked between these two dots.

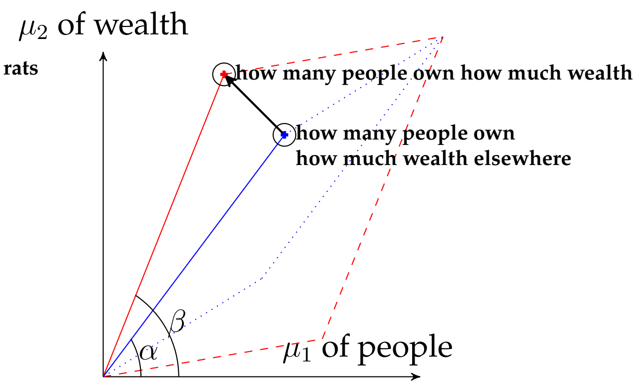

Figure 6: *Fewer people own more of the overall wealth here than elsewhere* (see (54))
The two high-lighted dots represent the cases of maximal informativeness at the matrix and *than*-clause level: maximal informativeness involves maximal ratio between measurements of two dimensions.
The inequalities along each dimension are checked between these two dots.

When both comparative expressions share the same polarity (e.g., *more* …*more*), they contribute to the QUD-based informativeness in a parallel way, evoking a QUD that integrates multi-dimensional information in a parallel way (see (53) and Fig. 5). When the two comparatives differ in polarity, they contribute in opposite ways, typically indicating a ratio-based computation of informativeness (see (54) and Fig. 6).

Once the computation of QUD-based maximal informativeness is established, the maximal informative cases at both the matrix and *than*-clause level can be identified (see the highlighted dots in Figs. 5 and 6). Based on these cases, comparisons are conducted between their measurements along each dimension. In this sense, comparisons play the same role as the cardinality tests in cumulative reading sentences (see §3.1).

For a sentence like (53) (see Fig. 5), increases along both dimensions ensure that the matrix clause expresses a higher level of QUD-based informativeness (e.g., collective success in hunting) than the *than*-clause. For a sentence like (54) (see Fig. 6), a decrease in population and an increase in wealth owned together make the matrix clause reach higher informativeness (e.g., the degree of wealth inequality) than the *than*-clause does.

The analysis outlined here accounts for all the key observations presented in §2. In a multi-head comparative, both the matrix and *than*-clause provide multi-dimensional measurements, supporting comparisons along multiple dimensions (see the discussion on multiple comparisons and parallelism in §2.1 and §2.2.4). These comparisons jointly contribute to the increase of informativeness regarding a single context-dependent issue (see §2.2.6). Whether the comparisons have the same or opposite directions (see §2.2.1) reflects the nature of this issue and how informativeness is calculated from the measurement values.[8] Based on these ideas, §4 will present a formal analysis.

4 Formal analysis of multi-head comparatives

The formal analysis of a multi-head comparative (§4.3) builds on two components: (i) a dynamic semantics implementation of maximizing informativeness (§4.1), and (ii) an additivity-based view of comparatives (§4.2).

4.1 The meaning of a cumulative-reading sentence

Following Brasoveanu (2013), I derive the meaning of cumulative-reading sentences within Dynamic Predicate Logic (DPL, Groenendijk and Stokhof 1991).

Within dynamic semantics, meaning derivation is considered a series of updates from an information state to another. Given the distributivity of DPL (see (55b), Groenendijk and Stokhof 1990), an update is a relation between assignment functions (i.e., it takes an assignment function as input and returns a set of assignment functions).

(55) DPL (see Groenendijk and Stokhof 1990, 1991):

a. **Information state** i: a set of assignment functions g Type: $\langle st \rangle$

b. **Update** r: from an information state to another Type: $\langle st, st \rangle$
Given that for every information state i, $r(i) = \bigcup_{g \in i} r(\{g\})$, an update can be considered a relation between assignment functions (of type $\langle s, st \rangle$).

c. **Truth**: an update is true if it yields a non-empty set.

According to Brasoveanu (2013) (see §3.1), modified numerals make semantic contribution in several layers: (i) introducing drefs, which further get restrictions like $\textsc{movie}(y), \textsc{boy}(x), \textsc{saw}(x, y)$ (see (56a)); (ii) imposing mereology-based maximality

[8] The rest of the discussion in §2 is about how multi-head comparatives are similar to regular comparatives in (i) being flexible with measurement units (§2.2.2), (ii) supporting measurements / comparisons beyond quantity (§2.2.3), and (iii) not necessarily requiring the overt presence of a *than*-clause (§2.2.5).

operators $\mathbf{M}_{u,\nu}$, so that the maximal boy-sum and movie-sum that satisfy all the relevant restrictions are picked out (see (56b)); (iii) imposing cardinality tests $\mathbf{3}_u$ and $\mathbf{5}_\nu$ (see (56c)). The cumulative reading of (42) is an update such that it is true if the cardinality of all boys who saw movies is 3 and the cardinality of all movies seen by boys is 5.

(56) $[\![$Exactly three^u boys saw exactly five$^\nu$ movies$]\!]$ (=(42))

$$\Leftrightarrow \underbrace{\mathbf{3}_u[\mathbf{5}_\nu}_{\text{cardinality tests}} \; [\; \underbrace{\mathbf{M}_{u,\nu}}_{\text{maximality}} \; \underbrace{[\![\text{some}^u \text{ boys saw some}^\nu \text{ movies}]\!]]]}_{\text{dref introduction}}$$

a. **Introducing drefs**: $[\![$some^u boys saw some$^\nu$ movies$]\!]$

$$\Leftrightarrow \lambda g.\left\{ g^{u\mapsto x}_{\nu\mapsto y} \,\middle|\, \textsc{movie}(y), \textsc{boy}(x), \textsc{saw}(x,y) \right\}$$

b. **Simultaneously applying mereology-based maximality operators**:

$$\mathbf{M}_{u,\nu} \overset{\text{def}}{=} \lambda m_{\langle s,st\rangle}.\lambda g_s.\{h \in m(g) \mid \neg\exists h' \in m(g).h(u) \sqsubset h'(u) \vee h(\nu) \sqsubset h'(\nu)\}$$

Thus $\mathbf{M}_{u,\nu}[\![$some^u boys saw some$^\nu$ movies$]\!]$

$$\Leftrightarrow \mathbf{M}_{u,\nu}[\lambda g.\left\{ g^{u\mapsto x}_{\nu\mapsto y} \,\middle|\, \textsc{movie}(y), \textsc{boy}(x), \textsc{saw}(x,y) \right\}]$$

$$\Leftrightarrow \lambda g.\left\{ g^{u\mapsto x}_{\nu\mapsto y} \,\middle|\, y = \sigma y[\textsc{movie}(y) \wedge \textsc{saw}(x,y)], x = \sigma x[\textsc{boy}(x) \wedge \textsc{saw}(x,y)] \right\}$$

c. **Checking cardinalities**:

$$\mathbf{3}_u \overset{\text{def}}{=} \lambda m_{\langle s,st\rangle}.\lambda g_s.\begin{cases} m(g) & \text{if } |g(u)| = 3 \\ \varnothing & \text{otherwise} \end{cases}, \; \mathbf{5}_\nu \overset{\text{def}}{=} \lambda m_{\langle s,st\rangle}.\lambda g_s.\begin{cases} m(g) & \text{if } |g(\nu)| = 5 \\ \varnothing & \text{otherwise} \end{cases}$$

Thus $\mathbf{3}_u[\mathbf{5}_\nu[\mathbf{M}_{u,\nu}[\![$some^u boys saw some$^\nu$ movies$]\!]]]$

$$\Leftrightarrow \lambda g.\left\{ g^{u\mapsto x}_{\nu\mapsto y} \,\middle|\, \begin{array}{l} y = \sigma y[\textsc{movie}(y) \wedge \textsc{saw}(x,y)] \\ x = \sigma x[\textsc{boy}(x) \wedge \textsc{saw}(x,y)] \end{array} \right\}, \text{ if } |x| = 3 \wedge |y| = 5$$

The discussion in §3.2 shows that Brasoveanu (2013)'s simultaneous application of two mereology-based maximality operators is a special case of a degree-QUD-based maximality of informativeness (see Fig. 3). Thus I define a generalized, degree-QUD-based maximality operator, which selects, among all drefs (assigned to $u_1, u_2, \ldots$), those that lead to maximal informativeness in resolving a degree QUD, G_{QUD}.

(57)
$$\mathbf{M}_{\text{QUD-INFO}} \overset{\text{def}}{=} \lambda m_{\langle s,st\rangle}.\lambda g_s.\{h \in m(g) \mid \neg\exists h' \in m(g).$$
$$G_{\text{QUD}}(\langle \mu_{11}(h'(u_1)), \mu_{12}(h'(u_1)), \ldots \mu_{21}(h'(u_2)), \ldots\rangle) >$$
$$G_{\text{QUD}}(\langle \mu_{11}(h(u_1)), \mu_{12}(h(u_1)), \ldots \mu_{21}(h(u_2)), \ldots\rangle)\}$$

(i.e., $\mathbf{M}_{\text{QUD-INFO}}$ picks out the assignment function h that assigns to $u_1, u_2, \ldots$ the drefs that resolve a degree QUD G_{QUD} with maximal informativeness)

In (57), h, h' are assignment functions; $h(u_i), h(u_j), \ldots$ are drefs assigned to $u_i, u_j, \ldots$; μ_{ij} (of type $\langle ed \rangle$) denotes a measure function, mapping the dref assigned to u_i to a degree along a relevant scale. The same dref can be potentially measured by multiple measure functions along distinct dimensions (e.g., μ_{11} and μ_{12} can both measure $h(u_1)$).

$\mathbf{M}_{\text{QUD-INFO}}$ includes a measure function G_{QUD}, which takes a tuple of measurements of drefs and returns a degree representing informativeness. G_{QUD} integrates information from multiple dimensions into a single value: its (non-curried) type is $\langle\langle d, d, \ldots\rangle, d\rangle$.

In (56), *exactly three*^{u} *boys saw exactly five*$^\nu$ *movies*, G_{QUD} takes two cardinalities as inputs and is monotonically increasing in both arguments (see (58)). The application of $\mathbf{M}_{\text{QUD-INFO}}$ selects the maximal boy- and movie-sums involved in seeing events:

(58) $$G_{\text{QUD}} \text{ is such that } \begin{cases} \forall d_1, d_2, d_1'[d_1 \leq d_1' \rightarrow G_{\text{QUD}}(\langle d_1, d_2\rangle) \leq G_{\text{QUD}}(\langle d_1', d_2\rangle)] \text{ and} \\ \forall d_1, d_2, d_2'[d_2 \leq d_2' \rightarrow G_{\text{QUD}}(\langle d_1, d_2\rangle) \leq G_{\text{QUD}}(\langle d_1, d_2'\rangle)] \end{cases}$$

In Krifka (1999)'s Guatemala example (repeated as (59), see §3.2 and Fig. 4), modified numerals also contribute in several layers: (i) introducing drefs, which further get restrictions like $\textsc{people}(x)$, $\textsc{land}(y)$, $\textsc{own}(x, y)$; (ii) imposing $\text{M}_{\text{QUD-INFO}}$; (iii) imposing tests **at-most-3%**$_u$ and **at-least-70%**$_\nu$. Crucially, this sentence is most felicitously uttered in a context where $\mathbf{M}_{\text{QUD-INFO}}$ selects the drefs that yield the largest ratio between the two measurements, i.e., the maximal informativeness in measuring, say, wealth imbalance.

(59) At most 3%^{u} of the population own at least 70%$^\nu$ of the land. (=(47))

$$\Leftrightarrow \underbrace{\textbf{at-most-3\%}_u[\textbf{at-least-70\%}_\nu}_{\text{tests}}[\mathbf{M}_{\text{QUD-INFO}}\ \underbrace{[\![\text{some}^u \text{ people own some}^\nu \text{ land}]\!]}_{\text{dref introduction}}]]$$

$$\Leftrightarrow$$

$$\lambda g\,.\left\{ g^{\substack{\nu\mapsto y \\ u\mapsto x}} \;\middle|\; \begin{array}{l} y = \sigma y[\textsc{land}(y) \wedge \textsc{own}(x, y) \wedge G_{\text{QUD}}(\langle \mu_{\text{people}}(x), \mu_{\text{land}}(y)\rangle) \text{ is maximal}] \\ x = \sigma x[\textsc{people}(x) \wedge \textsc{own}(x, y) \wedge G_{\text{QUD}}(\langle \mu_{\text{people}}(x), \mu_{\text{land}}(y)\rangle) \text{ is maximal}] \end{array} \right\},$$

if $\mu_{\text{people}}(x) \leq 3\% \wedge \mu_{\text{land}}(y) \geq 70\%$

$$\text{Here } G_{\text{QUD}} \text{ is such that } \begin{cases} \forall d_1, d_2, d_1'[d_1 \leq d_1' \rightarrow G_{\text{QUD}}(\langle d_1', d_2\rangle) \leq G_{\text{QUD}}(\langle d_1, d_2\rangle)] \text{ and} \\ \forall d_1, d_2, d_2'[d_2 \leq d_2' \rightarrow G_{\text{QUD}}(\langle d_1, d_2\rangle) \leq G_{\text{QUD}}(\langle d_1, d_2'\rangle)] \end{cases}$$

(i.e., G_{QUD} is monotonically decreasing along the population dimension, but monotonically increasing along the amount of land owned, e.g., $G_{\text{QUD}} = \lambda d_1.\lambda d_2.\frac{d_2}{d_1}$)

4.2 An additivity-based view of comparatives

4.2.1 Comparative morphemes as additive particles

According to the canonical view (e.g., von Stechow 1984, Heim 1985, Kennedy 1999, Schwarzschild 2008, Beck 2011), a comparative expresses an inequality (see (60a)).

Following Zhang and Ling (2021), Zhang and Zhang (2024), I adopt a new, additivity-based view: (i) English comparative morpheme *-er*/*more* is an additive particle in the domain of degrees, denoting an increase anaphoric to a base item, and (ii) a comparative expresses an increase or decrease along a relevant scale (see (60b)).

(60) Lucy is taller than Mary is ~~tall~~. **Comparative**

a. **Canonical, inequality-based view**: Lucy's height > Mary's height

b. **Additivity-based view**: Along the height scale, the position corresponding to Lucy's height degree involves an **increase** (i.e., a distance) from the **base position** corresponding to Mary's height

In English, *-er*/*more* is parallel to additive particle *another* in their **additive use**, as discussed in Greenberg (2010) and Thomas (2010, 2011).

(61) **Parallelism between *another* and additive *more* (in a domain of entities)**

a. I ate $\underbrace{\text{an apple}}_{\text{the base}}$. Then I ate $\underbrace{\textbf{another}}_{\text{an increase}}$.

b. I ate $\underbrace{\text{two bars of chocolate}}_{\text{the base}}$. Then I ate $\underbrace{\text{(a bit) }\textbf{more}\text{ (chocalate)}}_{\text{an increase}}$.

In (61), both *another* (see (61a)) and *more* (see (61b)) denote an increase over a base item in a **domain of entities**. In (61b), the amount of (*a bit*) *more* (*chocolate*) does not necessarily exceed the previously mentioned amount, *two bars*, indicating that here a non-comparative interpretation of *more* exists: this is the additive use of *more*.

The **comparative use of *-er*/*more*** can likewise be considered an increase over a base item, in a **domain of degrees**. In (62), the **base position** is Mary's height, and *-er* denotes an **increase** from this base position, resulting in a higher degree. This increase can be further measured, as illustrated by the numerical restriction 2 *inches* in (62a) and (62b).[9]

[9] The comparative use of *-er* in (62b) is also similar to additive particle *another* in (i) in two aspects: (i) in both cases, the base and the increase occur within the same sentence (cf. the cross-sentential use in (61) and (62a)); (ii) both cases involve a restriction on the increase (*2 inches* in (62b) and *Lucy* in (i)).

(62) **Comparative use: the increase and its base are in domains of degrees**

a. $\underbrace{\text{Mary is tall}}_{\text{the base: Mary's height}}$. Sue is $\underbrace{\text{(2 inches)}}_{\text{the restriction of the increase}}$ tall $\underbrace{\textbf{er}}_{\text{an increase based on Mary's height}}$.

b. Sue is $\underbrace{\text{(2 inches)}}_{\text{the restriction of the increase}}$ tall $\underbrace{\textbf{er}}_{\text{an increase}}$ $\underbrace{\text{than Mary is } \cancel{\text{tall}}}_{\text{the base: Mary's height}}$.

Thus, *-er*/*more* can be considered a cross-domain additive particle, denoting an increase in a domain of entities (in its additive use) or degrees (in its comparative use).[10]

In comparatives, our intuitions about the base for *-er*/*more* reveal two distinct semantic contributions of gradable adjectives: (i) **introducing a degree dref** (i.e., a **measurement** position on a scale) and relating it to an entity via a measure function (see Kennedy 1999) and (ii) indicating the **direction** towards higher informativeness.

1. Introducing a degree dref. In interpreting (63)–(66), intuitively, the base for *-er*/*less* is about Mary's position on a height scale, not about a numerical value like *5 feet* (see (63)) or a positive threshold like $\text{POS}_{\text{short}}$ / POS_{tall} (see (64)–(66); see also Li 2023).

Evidently, gradable adjective *tall* or *short* (underlined in (63)–(66)) must play a role in introducing a degree dref related to Mary, which serves as a base position supporting the increase or decrease denoted by cross-sentential anaphora *-er*/*less*.

In contrast, *5 feet* in (63) and $\text{POS}_{\text{short}}$ / POS_{tall} in (64)–(66) lack the status of a salient degree dref, failing to provide a base position that supports anaphoric *-er*/*less*.[11]

(i) $\underbrace{\text{A girl, Mary}}_{\text{the base and its restriction}}$, met $\underbrace{\textbf{another}\text{ girl, Lucy}}_{\text{an increase and its restriction}}$.

Further evidence for the parallel between the comparative and additive uses of *-er*/*more* and *another* comes from their ability to appear repetitively, expressing cumulative increases (Zhang and Zhang 2024):

(ii) **Repetitive use of *-er*/*more* and *another*: the accumulation of increases**

a. Lucy is tall<u>er</u> and tall<u>er</u>. **Comparative**

b. I bought <u>more</u> and <u>more</u> books. **Additive**

c. Janice had a little lamb and <u>another</u> and <u>another</u> and <u>another</u>. **Additive**

[10] In a domain of entities, both the base and the increase are of the same things (e.g., apples in (61a), chocolate in (61b)). However, in a domain of degrees (see (62)), the **base** means a **position** on a scale, while an **increase** is a **distance** away from that position. Positions (i.e., measurements) and distances between positions (i.e., differences between measurements) are conceptually distinct (see Zhang and Ling 2021).

[11] According to Zhang and Zhang (2024), *5 feet* in (63) denotes the **distance** between Mary's height and the absolute zero point on the height scale. Thus, as a distance, *5 feet* fails to serve as a base **position** for anaphoric *-er*. The case of *Lucy is taller than 5 feet* is different: *5 feet* denotes a position on the height scale. On the other hand, a contextual positive threshold, $\text{POS}_{\text{short}}$ / POS_{tall}, is probably too vague to be a dref.

(63) Mary is not even 5 feet tall. Lucy is **taller**. ⊭Lucy is taller than 5 feet

(64) Mary is short. Lucy is **taller**. ⊭ Lucy is not short (i.e., Lucy is taller than $\text{POS}_{\text{short}}$)

(65) Mary is tall. Lucy is **shorter**. ⊭ Lucy is not tall (i.e., Lucy is shorter than POS_{tall})

(66) Mary is tall. Lucy is **less tall**. ⊭ Lucy is not tall (i.e., Lucy is less tall than POS_{tall})

2. Indicating the direction towards higher informativeness. In supporting cross-sentential anaphora *-er*/*less* that expresses a change, the base position needs to be interpreted from a perspective that reflects the direction towards higher informativeness.

(67) a. Lucy is taller than the boys. ⟦than the boys⟧ ⇝ than the tallest boy
b. Lucy is shorter than the boys. ⟦than the boys⟧ ⇝ than the shortest boy
c. Lucy is less tall than the boys. ⟦than the boys⟧ ⇝ than the shortest boy

In (67a), *taller* addresses 'tallness reached', associating a higher height to higher informativeness. Thus, our intuition is that the base amounts to the highest degree that the boys can be as tall as, not the lowest degree that the boys can be as short as. The use of *-er* further extends the base to a new, higher value and thus, higher informativeness.

In (67b) and (67c), *shorter* or *less tall* addresses 'shortness reached' or 'tallness not reached', associating a lower height to higher informativeness. Thus, our intuition is that the base amounts to the lowest degree that the boys can be as short as, not the highest degree that the boys can be as tall as. With the use of (*short*)*er* or *less* (*tall*), the change further extends the base to a new, lower value and thus, higher informativeness.

This collaboration between the direction of a gradable adjective (*tall* vs. *short*) and the use of *-er*/*less* (in expressing a change and thus extending informativeness) aligns with the view of Beaver and Clark (2009) (see also Thomas 2011): **additivity** exhibits **QUD-based anaphoricity** (see also Roberts 1996/2012, Büring 2003, Zeevat 2004, Zeevat and Jasinskaja 2007). Additive particles (e.g., *another*, *also*) are anaphoric to a contextually salient QUD, indicating an extension of a salient partial answer in addressing the Current Question.

Under this view, though comparatives in (68) share the same truth conditions (see also Fig. 7), they address different degree QUDs.[12] Their *than*-clause is a partial answer

[12]Some QUDs are more natural and less marked than others, making some comparatives sound more natural than their truth-conditionally equivalent counterparts. E.g., out of the blue, *Lucy is taller than Mary is* is more acceptable than *Lucy is less short than Mary is*. Human cognition seems to treat the two opposing

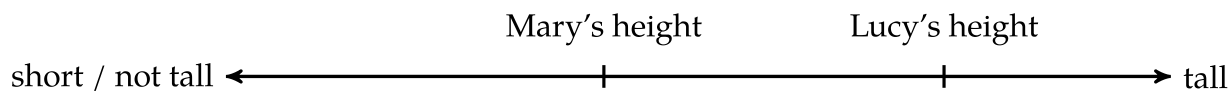

Figure 7: For a QUD about tallness reached (e.g., (68a)), the increase of height leads to higher informativeness. For a QUD about shortness reached (e.g., (68b)) or tallness not reached (e.g., (68c)), the decrease of height leads to higher informativeness.

to their current degree QUD at the matrix level. By including an additive particle that extends the base in addressing the matrix-level QUD, the matrix clause is always associated with higher informativeness over the base – the *than*-clause.[13]

(68) a. Lucy is taller than Mary is ~~tall~~. *-er* extends 'tallness reached by Mary' to address 'tallness reached by Lucy'

b. Mary is shorter than Lucy is ~~short~~. *-er* extends 'shortness reached by Lucy' to address 'shortness reached by Mary'

c. Mary is less tall than Lucy is. *less* extends 'tallness not reached by Lucy' to address 'tallness not reached by Mary'

4.2.2 Formal analysis of comparatives

I use dynamic semantics (particularly the informativeness-based maximality operator from §4.1) with the additivity-based view to develop a formal analysis of comparatives.

Under the canonical view, as illustrated in (69) and (70), a gradable adjective relates a degree d and an individual x, meaning that, along a relevant scale, the measurement of x reaches (i.e., is tall or short to) degree d (see e.g., Beck 2011 for a detailed review).

(69) $[\![\text{tall}]\!] \overset{\text{def}}{=} \lambda d.\lambda x.\textsc{height}(x) \geq d$

(70) $[\![\text{short}]\!] \overset{\text{def}}{=} \lambda d.\lambda x.\textsc{height}(x) \leq d$

Based on the discussion in §4.2.1, I make a distinction between two semantic

directions of a scale in an asymmetric way. This issue is probably related to our world knowledge and goes beyond the scope of the current paper. I thank an anonymous reviewer for raising this point.

[13] The current 'partial answer' view on the *than*-clause aligns with some existing analyses: Fleisher (2018, 2020) claims that a *than*-clause corresponds to a degree question; Zhang and Ling (2021) claims that a *than*-clause denotes the maximally informative true answer to a corresponding degree question, with *than* working as an answerhood operator à la Dayal (1996) (In English, word-initial [ð] correlates with definiteness). The current discussion adopts Beaver and Clark (2009)'s view on additivity and thus explains why this *than*-clause-level degree question must be parallel to (i.e., in the same direction as) the matrix-level QUD.

contributions of a gradable adjective: (i) introducing a degree dref and relating it to an entity (see (69) and (70)), and (ii) contributing to G_{QUD} for maximizing informativeness.

Just like modified numerals in cumulative-reading sentences, a comparative expression like *taller* in (71) makes contribution in three layers:

(71) Lucy is taller^{u_1} than Mary is ~~tall~~^{u_2}. (=(68a), see Fig. 7)

$$\Leftrightarrow \underbrace{\textbf{-er}_{u_1,u_2}}_{\text{3. tests}}[\ \underbrace{\mathbf{M}_{\text{QUD-INFO}}}_{\text{2. maximization}}\ \underbrace{[\![\text{Lucy is tall to a height } u_1, \text{Mary is tall to a height } u_2]\!]}_{\text{1. dref introduction}}]$$

First, within *taller*, gradable adjective *tall* introduces a degree dref and relates it to an entity via a measure function (as shown in (69)). Given that there is a parallel between the matrix and *than*-clause (see §2.2.4), we assume an elided *tall* within the *than*-clause.

(72) **Introducing degree drefs**: $[\![\text{Lucy is tall to } u_1, \text{Mary is tall to a } u_2]\!]$

$$\Leftrightarrow \lambda g.\left\{ g^{u_1\mapsto d_1}_{u_2\mapsto d_2} \,\middle|\, d_1 \leq \textsc{height}(\text{Lucy}), d_2 \leq \textsc{height}(\text{Mary}) \right\}$$

Second, by indicating the direction towards higher informativeness, *tall* also contributes to G_{QUD} in the application of the informativeness maximization operator $\mathbf{M}_{\text{QUD-INFO}}$. Thus, in this case, G_{QUD} takes a height measurement as input and is a monotonically increasing function. $\mathbf{M}_{\text{QUD-INFO}}$ picks out the drefs corresponding to the highest degrees (i.e., with maximal informativeness) at the matrix and *than*-clause level.

(73) **Applying $\mathbf{M}_{\text{QUD-INFO}}$ at both the *than*-clause and matrix level**:

$$\mathbf{M}_{\text{QUD-INFO}} \overset{\text{def}}{=} \lambda m.\lambda g.\{h \in m(g) \mid \neg\exists h' \in m(g).$$
$$\underbrace{G_{\text{QUD}}(h'(u_1)) > G_{\text{QUD}}(h(u_1))}_{\text{at the matrix level}} \vee \underbrace{G_{\text{QUD}}(h'(u_2)) > G_{\text{QUD}}(h(u_2))}_{\text{at the } \textit{than}\text{-clause level}}\}$$

Here the degree QUD is about **tallness reached** (see Fig. 7), i.e.,

G_{QUD} is such that $\forall d, d'[d \leq d' \rightarrow G(d) \leq G(d')]$ (i.e., monotonically increasing)

Thus $\mathbf{M}_{\text{QUD-INFO}}[\![\text{Lucy is tall to } u_1, \text{Mary is tall to } u_2]\!]$

$$\Leftrightarrow \mathbf{M}_{\text{QUD-INFO}}[\lambda g.\left\{ g^{u_1\mapsto d_1}_{u_2\mapsto d_2} \,\middle|\, d_1 \leq \textsc{height}(\text{Lucy}), d_2 \leq \textsc{height}(\text{Mary}) \right\}]$$

$$\Leftrightarrow \lambda g.\left\{ g^{u_1\mapsto d_1}_{u_2\mapsto d_2} \,\middle|\, d_1 = \textsc{height}(\text{Lucy}), d_2 = \textsc{height}(\text{Mary}) \right\}$$

Finally, within *taller*, the additive particle *-er* checks whether there is an increase of informativeness between the two maximally informative drefs selected by $\mathbf{M}_{\text{QUD-INFO}}$:

(74) **Checking the increase of informativeness along the relevant scale**:

$$\mathbf{er}_{u_1,u_2} \overset{\text{def}}{=} \lambda m.\lambda g.\begin{cases} m(g) & \text{if } G_{\text{QUD}}(g(u_1)) > G_{\text{QUD}}(g(u_2)) \\ \varnothing & \text{otherwise}\end{cases}$$

Thus $\textbf{-er}_{u_1,u_2}[\mathbf{M}_{\text{QUD-INFO}}[\![\text{Lucy is tall to } u_1, \text{Mary is tall to } u_2]\!]]$

$\Leftrightarrow \lambda g.\left\{ g^{\substack{u_2\mapsto d_2\\ u_1\mapsto d_1}} \middle| d_2 = \textsc{height}(\text{Mary}), d_1 = \textsc{height}(\text{Lucy}) \right\}$, if $d_1 > d_2$

A *shorter-than* or *less-tall-than* comparative (see (75)) can be derived in the same way as a *taller-than* comparative. Intuitively, *shorter-than* and *less-tall-than* comparatives have the same truth conditions. To capture this, I decompose *less* into two parts: a type-shifter called LITTLE that changes a gradable adjective into its antonym, and additive particle *-er*.

(75) a. Mary is shorter^{u_1} than Lucy is ~~short^{u_2}~~. (=(68b))
b. Mary is less tall^{u_1} than Lucy is . (=(68c))
$\rightsquigarrow$ Mary lacks more height^{u_1} than Lucy does ~~lack height^{u_2}~~.

(76) a. $\textsc{little} \overset{\text{def}}{=} \lambda P_{\langle d,\langle et\rangle\rangle}.\lambda d.\lambda x.\begin{cases} P\text{-}\textsc{measure}(x) \leq d & \text{if } P \overset{\text{def}}{=} \lambda d.\lambda x.P\text{-}\textsc{measure}(x) \geq d \\ P\text{-}\textsc{measure}(x) \geq d & \text{if } P \overset{\text{def}}{=} \lambda d.\lambda x.P\text{-}\textsc{measure}(x) \leq d \end{cases}$
(P-MEASURE is the measure function associated with gradable adjective P)
b. $\textsc{little}[\![\text{tall}]\!] = \lambda d.\lambda x.\textsc{height}(x) \leq d = [\![\text{short}]\!]$

Thus, in deriving the meaning for the sentences in (75), within *shorter* or *less tall*, *short* or LITTLE($[\![$tall$]\!]$) introduces a degree dref related to an entity via a measure function.

(77) **Introducing degree drefs**: $[\![\text{Mary is short to } u_1, \text{Lucy is short to } u_2]\!]$
$\Leftrightarrow \lambda g.\left\{ g^{\substack{u_2\mapsto d_2\\ u_1\mapsto d_1}} \middle| \textsc{height}(\text{Mary}) \leq d_1, \textsc{height}(\text{Lucy}) \leq d_2 \right\}$

Then *short* or LITTLE($[\![$tall$]\!]$) also indicates the direction towards higher informativeness. Specifically, in implementing $\mathbf{M}_{\text{QUD-INFO}}$, the function G_{QUD} is monotonically decreasing, so that $\mathbf{M}_{\text{QUD-INFO}}$ picks out the drefs corresponding to the lowest heights (i.e., with maximal informativeness) at the matrix and *than*-clause level.

(78) The degree QUD is about **shortness reached** or **tallness not reached** (see Fig. 7): G_{QUD} is s.t. $\forall d, d'[d \leq d' \to G(d') \leq G(d)]$, i.e., monotonically decreasing

Finally, within *shorter* or *less tall*, additive particle *-er* checks whether there is an increase of informativeness between the two maximally informative drefs selected by $\mathbf{M}_{\text{QUD-INFO}}$. Since G_{QUD} is monotonically decreasing, $G_{\text{QUD}}(d_1) > G_{\text{QUD}}(d_2)$ amounts to $d_1 < d_2$.

(79) $\textbf{-er}_{u_1,u_2}[\mathbf{M}_{\text{QUD-INFO}}[\![\text{Mary is short to } u_1\text{, Lucy is short to } u_2]\!]]$

$$\Leftrightarrow \lambda g.\left\{g^{u_1\mapsto d_1}_{u_2\mapsto d_2}\;\middle|\;\begin{array}{l} d_2 = \text{the lowest } d \text{ s.t. } \textsc{height}(\text{Lucy}) \leq d \\ d_1 = \text{the lowest } d \text{ s.t. } \textsc{height}(\text{Mary}) \leq d\end{array}\right\},$$

if $G_{\text{QUD}}(g(u_1)) > G_{\text{QUD}}(g(u_2))$

$$\Leftrightarrow \lambda g.\left\{g^{u_1\mapsto d_1}_{u_2\mapsto d_2}\;\middle|\; d_2 = \textsc{height}(\text{Lucy}), d_1 = \textsc{height}(\text{Mary})\right\}, \text{ if } d_1 < d_2$$

For these regular comparatives, the derived truth conditions shown in (74) and (79) are exactly the same as those derived in the canonical approach to comparatives: i.e., $\textsc{height}(\text{Lucy}) > \textsc{height}(\text{Mary})$, and $\textsc{height}(\text{Mary}) < \textsc{height}(\text{Lucy})$. In §5.1, I will show further empirical advantages of the current approach in analyzing sentences like *Lucy is taller than the boys* (see (67a)-(67c)).

4.3 The meaning of a multi-head comparative

Based on the analysis of cumulative-reading sentences (see §4.1) and comparatives (see §4.2), the meaning of a multi-head comparative is derived in three steps:

1. Introducing drefs (see (80a) and (81a)) Just as in the analysis of cumulatives, there are potentially plural **entity drefs**, and just as in the analysis of comparatives, there are **degree drefs**.[14] Restrictions are added onto these entity and degree drefs.

2. Applying $\mathbf{M}_{\text{QUD-INFO}}$ (see (80b) and (81b)) This informativeness maximization operator selects the drefs that yield maximal informativeness in addressing the contextually salient degree QUD – at the matrix and *than*-clause level.

For (80b), the G_{QUD} involved in $\mathbf{M}_{\text{QUD-INFO}}$ is monotonically increasing along both dimensions, thus selecting the upper-rightmost dots highlighted in Fig. 5 along with their measurements. For (81b), the G_{QUD} involved in $\mathbf{M}_{\text{QUD-INFO}}$ is monotonically decreasing and increasing along the dimensions of μ_{PEOPLE} and μ_{WEALTH}, respectively. Thus the two upper-leftmost dots highlighted in Fig. 6 along with their measurements are selected.

3. Applying the tests of additive particle *-er* (see (80c) and (81c)) Finally, for a pair of selected degree drefs that correspond to each other, **-er** checks whether there is an increase of informativeness. These tests are applied along each dimension. Thus in a

[14] In §4.2, to improve readability, I only include the drefs relevant to the current discussion: degree drefs.

context where 3% of the population own 70% of the wealth elsewhere, while 2% of the population own 70% of the wealth here, the sentence (81) is judged false, because along the dimension of μ_{WEALTH}, there is no increase of informativeness.[15] This is consistent with the view of von Stechow (1984): a multi-head comparative expresses two independent comparisons. I further show that what undergoes comparison needs to be based on an interplay between information along multiple dimensions, i.e., a single QUD.

(80) More dogs ate more rats than cats ate mice. (see Fig. 5)

⟦More^{u_1,m_1} dogs ate more$^{\nu_1,n_1}$ rats than ~~many^{u_2,m_2}~~ cats ate ~~many$^{\nu_2,n_2}$~~ mice⟧

$\Leftrightarrow$

-er$_{m_1,m_2}$[**-er**$_{n_1,n_2}$[$\mathbf{M}_{\text{QUD-INFO}}$⟦$m_1$-many^{u_1} dogs ate n_1-many$^{\nu_1}$ rats, m_2-many^{u_2} cats ate n_2-many$^{\nu_1}$ mice⟧]]

a. $\Leftrightarrow$

$$\textbf{-er}_{m_1,m_2}[\textbf{-er}_{n_1,n_2}[\mathbf{M}_{\text{QUD-INFO}}[\lambda g\,.\left\{ g^{\substack{m_1\mapsto d_{\text{AG1}}\\ n_1\mapsto d_{\text{THM1}}\\ \nu_1\mapsto y_1\\ u_1\mapsto x_1\\ m_2\mapsto d_{\text{AG2}}\\ n_2\mapsto d_{\text{THM2}}\\ \nu_2\mapsto y_2\\ u_2\mapsto x_2}} \;\middle|\; \begin{array}{l} \text{DOG}(x_1), \text{RAT}(y_1), \text{EAT}(x_1,y_1)\\ \text{CAT}(x_2), \text{MOUSE}(y_2), \text{EAT}(x_2,y_2)\\ d_{\text{AG1}} \leq \text{CARD}(x_1)\\ d_{\text{THM1}} \leq \text{CARD}(y_1)\\ d_{\text{AG2}} \leq \text{CARD}(x_2)\\ d_{\text{THM2}} \leq \text{CARD}(y_2) \end{array} \right\}]]]$$

b.
$$\Leftrightarrow \textbf{-er}_{m_1,m_2}[\textbf{-er}_{n_1,n_2}[\lambda g\,.\left\{ g^{\substack{m_1\mapsto d_{\text{AG1}}\\ n_1\mapsto d_{\text{THM1}}\\ \nu_1\mapsto y_1\\ u_1\mapsto x_1\\ m_2\mapsto d_{\text{AG2}}\\ n_2\mapsto d_{\text{THM2}}\\ \nu_2\mapsto y_2\\ u_2\mapsto x_2}} \;\middle|\; \begin{array}{l} x_1 = \sigma x_1[\text{DOG}(x_1) \wedge \text{EAT}(x_1,y_1)]\\ y_1 = \sigma y_1[\text{RAT}(y_1) \wedge \text{EAT}(x_1,y_1)]\\ x_2 = \sigma x_2[\text{CAT}(x_2) \wedge \text{EAT}(x_2,y_2)]\\ y_2 = \sigma y_2[\text{MOUSE}(y_2) \wedge \text{EAT}(x_2,y_2)]\\ d_{\text{AG1}} = \text{CARD}(x_1), d_{\text{THM1}} = \text{CARD}(y_1)\\ d_{\text{AG2}} = \text{CARD}(x_2), d_{\text{THM2}} = \text{CARD}(y_2) \end{array} \right\}]]$$

c.
$$\Leftrightarrow \lambda g\,.\left\{ g^{\substack{m_1\mapsto d_{\text{AG1}}\\ n_1\mapsto d_{\text{THM1}}\\ \nu_1\mapsto y_1\\ u_1\mapsto x_1\\ m_2\mapsto d_{\text{AG2}}\\ n_2\mapsto d_{\text{THM2}}\\ \nu_2\mapsto y_2\\ u_2\mapsto x_2}} \;\middle|\; \begin{array}{l} x_1 = \sigma x_1[\text{DOG}(x_1) \wedge \text{EAT}(x_1,y_1)]\\ y_1 = \sigma y_1[\text{RAT}(y_1) \wedge \text{EAT}(x_1,y_1)]\\ x_2 = \sigma x_2[\text{CAT}(x_2) \wedge \text{EAT}(x_2,y_2)]\\ y_2 = \sigma y_2[\text{MOUSE}(y_2) \wedge \text{EAT}(x_2,y_2)]\\ d_{\text{AG1}} = \text{CARD}(x_1), d_{\text{THM1}} = \text{CARD}(y_1)\\ d_{\text{AG2}} = \text{CARD}(x_2), d_{\text{THM2}} = \text{CARD}(y_2) \end{array} \right\},$$

$$\text{if} \begin{cases} \forall d[G_{\text{QUD}}(\langle d_{\text{AG1}}, d\rangle) > G_{\text{QUD}}(\langle d_{\text{AG2}}, d\rangle)] \\ \forall d[G_{\text{QUD}}(\langle d, d_{\text{THM1}}\rangle) > G_{\text{QUD}}(\langle d, d_{\text{THM2}}\rangle)] \end{cases} \quad \text{(i.e., } d_{\text{AG1}} > d_{\text{AG2}} \wedge d_{\text{THM1}} > d_{\text{THM2}}\text{)}$$

[15] I thank an anonymous reviewer for their insightful comment on this issue.

(81) Fewer people own more of the overall wealth here than elsewhere. (see Fig. 6)

$[\![\text{Fewer}^{u_1,m_1}\text{ people own more}^{\nu_1,n_1}\text{ wealth here than }\cancel{\text{few}^{u_2,m_2}}\text{ people own }\cancel{\text{much}^{\nu_2,n_2}}\text{ wealth elsewhere}]\!]$

$\Leftrightarrow$

$\textbf{-er}_{m_1,m_2}[\textbf{-er}_{n_1,n_2}[\mathbf{M}_{\text{QUD-INFO}}[\![m_1\text{-few}^{u_1}\text{ ppl-here own }n_1\text{-much}^{\nu_1}\text{ wth-here}, m_2\text{-few}^{u_2}\text{ ppl-else own }n_2\text{-much}^{\nu_1}\text{ wth-else}]\!]]]$

a. $\Leftrightarrow$

$$\textbf{-er}_{m_1,m_2}[\textbf{-er}_{n_1,n_2}[\mathbf{M}_{\text{QUD-INFO}}[\lambda g\,.\left\{ g^{\substack{m_1\mapsto d_{\text{PPL1}}\\ n_1\mapsto d_{\text{WTH1}}\\ \nu_1\mapsto y_1\\ u_1\mapsto x_1\\ m_2\mapsto d_{\text{PPL2}}\\ n_2\mapsto d_{\text{WTH2}}\\ \nu_2\mapsto y_2\\ u_2\mapsto x_2}} \;\middle|\; \begin{array}{l} \textsc{people-here}(x_1), \textsc{wealth-here}(y_1),\\ \textsc{own}(x_1,y_1)\\ \textsc{people-else}(x_2), \textsc{wealth-else}(y_2),\\ \textsc{own}(x_2,y_2)\\ d_{\text{PPL1}} \geq \mu_{\text{PEOPLE}}(x_1), d_{\text{WTH1}} \leq \mu_{\text{WEALTH}}(y_1)\\ d_{\text{PPL2}} \geq \mu_{\text{PEOPLE}}(x_2), d_{\text{WTH2}} \leq \mu_{\text{WEALTH}}(y_2) \end{array} \right\}]]]$$

b. $\Leftrightarrow$

$$\textbf{-er}_{m_1,m_2}[\textbf{-er}_{n_1,n_2}[\lambda g\,.\left\{ g^{\substack{m_1\mapsto d_{\text{PPL1}}\\ n_1\mapsto d_{\text{WTH1}}\\ \nu_1\mapsto y_1\\ u_1\mapsto x_1\\ m_2\mapsto d_{\text{PPL2}}\\ n_2\mapsto d_{\text{WTH2}}\\ \nu_2\mapsto y_2\\ u_2\mapsto x_2}} \;\middle|\; \begin{array}{l} x_1 = \sigma x_1[\textsc{people-here}(x_1) \wedge \textsc{own}(x_1,y_1)\wedge\\ G_{\text{QUD}}(\langle \mu_{\text{people}}(x_1), \mu_{\text{wealth}}(y_1)\rangle)\text{ is maximal}]\\ y_1 = \sigma y_1[\textsc{wealth-here}(y_1) \wedge \textsc{own}(x_1,y_1)\wedge\\ G_{\text{QUD}}(\langle \mu_{\text{people}}(x_1), \mu_{\text{wealth}}(y_1)\rangle)\text{ is maximal}]\\ x_2 = \sigma x_2[\textsc{people-else}(x_2) \wedge \textsc{own}(x_2,y_2)\wedge\\ G_{\text{QUD}}(\langle \mu_{\text{people}}(x_2), \mu_{\text{wealth}}(y_2)\rangle)\text{ is maximal}]\\ y_2 = \sigma y_2[\textsc{wealth-else}(y_2) \wedge \textsc{own}(x_2,y_2)\wedge\\ G_{\text{QUD}}(\langle \mu_{\text{people}}(x_2), \mu_{\text{wealth}}(y_2)\rangle)\text{ is maximal}]\\ d_{\text{PPL1}} = \mu_{\text{PEOPLE}}(x_1), d_{\text{WTH1}} = \mu_{\text{WEALTH}}(y_1)\\ d_{\text{PPL2}} = \mu_{\text{PEOPLE}}(x_2), d_{\text{WTH2}} = \mu_{\text{WEALTH}}(y_2) \end{array} \right\}]]$$

c. $$\Leftrightarrow \lambda g\,.\left\{ g^{\substack{m_1\mapsto d_{\text{PPL1}}\\ n_1\mapsto d_{\text{WTH1}}\\ \nu_1\mapsto y_1\\ u_1\mapsto x_1\\ m_2\mapsto d_{\text{PPL2}}\\ n_2\mapsto d_{\text{WTH2}}\\ \nu_2\mapsto y_2\\ u_2\mapsto x_2}} \;\middle|\; \begin{array}{l} x_1 = \sigma x_1[\textsc{people-here}(x_1) \wedge \textsc{own}(x_1,y_1)\wedge\\ G_{\text{QUD}}(\langle \mu_{\text{people}}(x_1), \mu_{\text{wealth}}(y_1)\rangle)\text{ is maximal}]\\ y_1 = \sigma y_1[\textsc{wealth-here}(y_1) \wedge \textsc{own}(x_1,y_1)\wedge\\ G_{\text{QUD}}(\langle \mu_{\text{people}}(x_1), \mu_{\text{wealth}}(y_1)\rangle)\text{ is maximal}]\\ x_2 = \sigma x_2[\textsc{people-else}(x_2) \wedge \textsc{own}(x_2,y_2)\wedge\\ G_{\text{QUD}}(\langle \mu_{\text{people}}(x_2), \mu_{\text{wealth}}(y_2)\rangle)\text{ is maximal}]\\ y_2 = \sigma y_2[\textsc{wealth-else}(y_2) \wedge \textsc{own}(x_2,y_2)\wedge\\ G_{\text{QUD}}(\langle \mu_{\text{people}}(x_2), \mu_{\text{wealth}}(y_2)\rangle)\text{ is maximal}]\\ d_{\text{PPL1}} = \mu_{\text{PEOPLE}}(x_1), d_{\text{WTH1}} = \mu_{\text{WEALTH}}(y_1)\\ d_{\text{PPL2}} = \mu_{\text{PEOPLE}}(x_2), d_{\text{WTH2}} = \mu_{\text{WEALTH}}(y_2) \end{array} \right\}]],$$

$$\text{if} \begin{cases} \forall d[G_{\text{QUD}}(\langle d_{\text{PPL1}}, d\rangle) > G_{\text{QUD}}(\langle d_{\text{PPL2}}, d\rangle)]\\ \forall d[G_{\text{QUD}}(\langle d, d_{\text{WTH1}}\rangle) > G_{\text{QUD}}(\langle d, d_{\text{WTH2}}\rangle)] \end{cases} \quad (\text{i.e., } d_{\text{PPL1}} < d_{\text{PPL2}} \wedge d_{\text{WTH1}} > d_{\text{WTH2}})$$

5 Discussion

The proposed analysis of multi-head comparatives offers further theoretical insights into the meaning of comparatives (see §5.1) and the notion of informativeness (see §5.2).

5.1 Comparing theories of comparatives

The analysis presented in §4.3 adopts the view that comparatives are a kind of additivity phenomenon and combines Dynamic Predicate Logic with degree semantics in formal implementation. Here I compare this approach with (i) canonical degree semantics and (ii) interval semantics, discussing how these theories differ in their analyses of some empirical data and highlighting the advantages of the current approach.

5.1.1 Accounting for regular comparatives

A comparative like (82) presents a challenge for the canonical degree semantics. As illustrated in (83), the derived truth condition is too weak: Mary is compared to the maximal height reached by all the boys, which is the height of the shortest boy. Intuitively, however, (82) means that Mary's height exceeds that of the tallest boy.

(82) Mary is taller than the boys ~~are tall~~. (=(67a))

(83) LF: -er [λd.Mary is d-tall][λd.the boys are DIST d-tall] **Degree semantics**

a. $[\![\text{tall}]\!] \overset{\text{def}}{=} \lambda d.\lambda x.\textsc{height}(x) \geq d$ (here x is an atomic individual) (= (69))

b. $[\![\text{-er}]\!] \overset{\text{def}}{=} \lambda D_{2\langle dt\rangle}, \lambda D_{1\langle dt\rangle}.\textsc{max}(D_1) > \textsc{max}(D_2)$
($\textsc{max} \overset{\text{def}}{=} \lambda D.\iota d[d \in D \wedge \forall d' \in D[d' \leq d]]$)

c. $[\![\lambda d.\text{ the boys are DIST } d\text{-tall}]\!] = \lambda d.\forall x \sqsubseteq_{\textsc{atom}} \sigma X(\textsc{boy}(X))[\textsc{height}(x) \geq d]$
(Here $\textsc{dist} \overset{\text{def}}{=} \lambda X.\lambda P.\forall x \sqsubseteq_{\textsc{atom}} X[P(x)]$)

d. $[\![(82)]\!] \Leftrightarrow \textsc{max}(\lambda d.\textsc{ht}(\text{Ma}) \geq d) > \textsc{max}(\lambda d.\forall x \sqsubseteq_{\textsc{atom}} \sigma X(\textsc{boy}(X))[\textsc{ht}(x) \geq d])$
$\Leftrightarrow \textsc{height}(\text{Mary}) > \textsc{height}(\text{the shortest boy})$

Interval semantics (see Zhang and Ling 2021, Zhang and Zhang 2024) can capture our intuitive interpretation of (82). An interval is a convex set of degrees.[16] Using an interval, we can represent either a precise or an imprecise position on a scale. For example, if Mary's height is measured with high precision, it can be represented as

[16] A set P is convex iff $\forall a, b \in P$ (say $a \leq b$), it follows that any x that is $a \leq x \leq b$ is also a member of P.

$[5'5'', 5'5'']$, i.e., an interval that is a singleton set of degrees. If the measurement is less precise, and Mary is estimated to be between $5'4''$ and $5'6''$, her height can be represented as $[5'4'', 5'6'']$, an interval standing for a not very precise position along the height scale.

A comparative is considered a relation among three intervals: two representing **positions** on a scale, and the third representing the **distance** (or difference) between them. In a *taller-than* comparative, the height interval associated with the matrix clause minus that associated with the *than*-clause yields a positive difference.

As shown in (84), a gradable adjective relates an interval I and an individual x, meaning that x's measurement falls within (i.e., is a subset of) I (see (84a)). The *than*-clause of (82), *than the boys ~~are tall~~*, denotes the narrowest interval that includes each boy's height, i.e., the interval ranging from the shortest to the tallest boy(s)' height (e.g., $[5'3'', 5'11'']$; see (84b)). The morpheme *-er* denotes the most general positive difference: the interval $(0, +\infty$ (see (84c)). A silent operator $\ominus$ takes two intervals, one as the difference and the other as the subtrahend, and returns the corresponding minuend interval (see (84d)). At the matrix level, the comparison target (here Mary) is associated with this **minuend** interval (see (84e)). Based on interval subtraction, the lower bound of the minuend is calculated from (i) the lower bound of the difference and (ii) **the upper bound of the subtrahend**. Thus our intuitive interpretation of (82) is derived.

(84) LF: Mary is tall $\ominus$[-er [than [λI.[the boys are DIST I-tall]]]] **Interval semantics**

a. $[\![\text{tall}]\!] \overset{\text{def}}{=} \lambda I_{\langle dt \rangle}.\lambda x.\textsc{height}(x) \subseteq I$ (here x is an atomic individual)

b. $[\![[\text{than } [\lambda I.[\text{the boys are } \textsc{dist } I\text{-tall}]]]]\!]$
$= [\![\text{than}]\!][\lambda I.\forall x \sqsubseteq_{\text{ATOM}} \sigma X(\textsc{boy}(X))[\textsc{height}(x) \subseteq I]]$
$\approx$ [the shortest boy(s)' height, the tallest boy(s)' height]
$([\![\text{than}]\!]_{\langle\langle dt,t\rangle,dt\rangle} \overset{\text{def}}{=} \lambda p_{\langle dt,t\rangle}.\iota I[p(I) \wedge \forall I'[p(I') \wedge I' \neq I \rightarrow I \subset I']])$

c. $[\![\text{-er}]\!] \overset{\text{def}}{=} (0, +\infty)$ (i.e., a default positive difference)

d. $\ominus \overset{\text{def}}{=} \lambda I_{\text{STDD}}.\lambda I_{\text{DIFF}}.\iota I[I - I_{\text{STDD}} = I_{\text{DIFF}}]$

e. $[\![(82)]\!] \Leftrightarrow \textsc{height}(\text{Mary}) \subseteq \iota I[I - [\text{shortest}, \text{tallest}] = (0, +\infty)]$
$\Leftrightarrow \textsc{height}(\text{Mary}) \subseteq (\text{the tallest boy(s)' height}, +\infty)$

In a *shorter-than* comparative, the lexical entry for *short* is considered the same as that of *tall* (see (85a)), but the use of *short* shifts the direction of comparison (see (85b)). At the matrix level, the comparison target is associated with the **subtrahend**, whose upper bound is calculated from (i) **the lower bound of the minuend** and (ii) the lower bound of the difference, yielding our intuitive interpretation.

(85) Mary is shorter than the boys ~~are short~~. (= (67b))

a. $[\![\text{short}]\!] \overset{\text{def}}{=} \lambda I_{\langle dt \rangle}.\lambda x.\textsc{height}(x) \subseteq I$ (here x is an atomic individual)

b. $[\![(85)]\!] \Leftrightarrow \textsc{height}(\text{Mary}) \subseteq \iota I[[\text{shortest}, \text{tallest}] - I = (0, +\infty)]$

$\Leftrightarrow \textsc{height}(\text{Mary}) \subseteq (-\infty, \text{the shortest boy(s)' height})$

Like interval semantics, the current dynamic degree semantics approach also derives the intuitive readings of these sentences. In (86) and (87), the application of $\mathbf{M}_{\text{QUD-INFO}}$ selects the drefs that yield maximal informativeness in addressing tallness or shortness reached. The comparison test checks whether informativeness increases from the *than*-clause to the matrix clause.

(86) $[\![\text{Mary is taller}^{u_1} \text{ than the}^{\nu} \text{ boys } \cancel{\text{are tall}^{u_2}}]\!]$

$\Leftrightarrow \textbf{-er}_{u_1,u_2}[\mathbf{M}_{\text{QUD-INFO}}[\![\text{Mary is } u_1\text{-tall, some}^{\nu} \text{ boys are } u_2\text{-tall}]\!]]$

$\Leftrightarrow \textbf{-er}_{u_1,u_2}[\mathbf{M}_{\text{QUD-INFO}}[\lambda g.\left\{ g^{\substack{u_1 \mapsto d_1 \\ u_2 \mapsto d_2 \\ \nu \mapsto x}} \middle| \begin{array}{l} \textsc{boy}(x), d_2 \leq \textsc{height}(x) \\ d_1 \leq \textsc{height}(\text{Mary}) \end{array} \right\}]]$

$\Leftrightarrow \lambda g.\left\{ g^{\substack{u_1 \mapsto d_1 \\ u_2 \mapsto d_2 \\ \nu \mapsto x}} \middle| \begin{array}{l} x = \text{the tallest boy} \\ d_2 = \textsc{height}(x), d_1 = \textsc{height}(\text{Ma}) \end{array} \right\}]$, if $G_{\text{QUD}}(g(u_1)) > G_{\text{QUD}}(g(u_2))$

(Here G_{QUD} is monotonically increasing, thus the test amounts to $d_1 > d_2$)

(87) $[\![\text{Mary is shorter}^{u_1} \text{ than the}^{\nu} \text{ boys } \cancel{\text{are short}^{u_2}}]\!]$

$\Leftrightarrow \textbf{-er}_{u_1,u_2}[\mathbf{M}_{\text{QUD-INFO}}[\![\text{Mary is } u_1\text{-short, some}^{\nu} \text{ boys are } u_2\text{-short}]\!]]$

$\Leftrightarrow \textbf{-er}_{u_1,u_2}[\mathbf{M}_{\text{QUD-INFO}}[\lambda g.\left\{ g^{\substack{u_1 \mapsto d_1 \\ u_2 \mapsto d_2 \\ \nu \mapsto x}} \middle| \begin{array}{l} \textsc{boy}(x), d_2 \geq \textsc{height}(x) \\ d_1 \geq \textsc{height}(\text{Mary}) \end{array} \right\}]]$

$\Leftrightarrow \lambda g.\left\{ g^{\substack{u_1 \mapsto d_1 \\ u_2 \mapsto d_2 \\ \nu \mapsto x}} \middle| \begin{array}{l} x = \text{the shortest boy} \\ d_2 = \textsc{height}(x), d_1 = \textsc{height}(\text{Ma}) \end{array} \right\}]$, if $G_{\text{QUD}}(g(u_1)) > G_{\text{QUD}}(g(u_2))$

(Here G_{QUD} is monotonically decreasing, thus the test amounts to $d_2 > d_1$)

5.1.2 Accounting for multi-head comparatives

The current approach is more advantageous than canonical degree semantics in analzying regular comparatives like (82). I have also shown that the interaction of multiple dimensions is essential for interpreting and analyzing multi-head comparatives, giving the current approach an edge over pure interval semantics as well.

For example, under interval semantics, (88) involves two independent comparisons (each represented as interval subtraction), one along the quality scale and the other along the price scale (see (89)).

(88) Slightly better products^{u_1} cost a lot more (than they^{u_2} did last year).
(adapted from https://shavercheck.com/best-electric-razor/)

(89) a. Comparison 1: $\mu_{\text{quality}}(g(u_1)) - \mu_{\text{quality}}(g(u_2)) \subseteq (0, d_1]$
(d_1: the threshold of being small for a difference of quality)

b. Comparison 2: $\mu_{\text{price}}(g(u_1)) - \mu_{\text{price}}(g(u_2)) \subseteq [d_2, +\infty)$
(d_2: the threshold of being large for a difference of price)

However, our intuitive interpretation of (88) also reflects a sense of disproportion: the value for money has decreased, due to the interplay between *slightly better* and *a lot more* (money). In other words, *slightly* and *a lot* together signal that the matrix clause conveys a lower value-for-money than the *than*-clause.

Thus G_{QUD}, which tracks 'how low value-for-money is', is monotonically increasing along μ_{PRICE}, but monotonically decreasing along μ_{QUALITY}. The comparison tests, **slightly-(bett)er** and **a-lot-more**, check the coupling between the two changes, and their overall effect (i.e., a decrease along the scale of 'value for money').[17]

(90) **slightly-(bett)er**$_{u_1,u_2}$, **a-lot-more**$_{u_1,u_2}$ together check whether
$\mu_{\text{quality}}(g(u_1)), \mu_{\text{quality}}(g(u_2)), \mu_{\text{price}}(g(u_1)), \mu_{\text{price}}(g(u_2))$ satisfy
$\underbrace{0 < \mu_{\text{quality}}(g(u_1)) - \mu_{\text{quality}}(g(u_2)) \leq d_1}_{\text{slightly better}} \wedge \underbrace{\mu_{\text{price}}(g(u_1)) - \mu_{\text{price}}(g(u_2)) \geq d_2}_{\text{a lot more}} \wedge$
$G(\langle \mu_{\text{price}}(g(u_1)), \mu_{\text{quality}}(g(u_1))\rangle) > G(\langle \mu_{\text{price}}(g(u_2)), \mu_{\text{quality}}(g(u_2))\rangle)$

5.2 A hidden degree QUD and informativeness

The current analysis of multi-head comparatives follows Krifka (1999) and Zhang (2023) in viewing informativeness as based on a hidden degree QUD. Are there other natural language phenomena involving a hidden degree QUD?

Zhang (2022) proposes that the presupposition of focus sensitive particle *even* is also based on a hidden degree QUD (see (92); see also Greenberg 2018 for a similar view).

Zhang (2022)'s view explains why an *even*-sentence may lack entity-based additivity or low likelihood in interpretation. Under the context of (91), no one other than Eeyore took a bite of the thistles, so the felicitous use of *even* does not require its prejacent to be

[17] Although I believe that the current dynamic degree semantics is advantageous and sufficient in accounting for various multi-head comparatives, interval semantics is independently motivated by data such as *Mary is between* $5'10''$ *and* $6'$ *tall* (analyzed as HEIGHT(Mary) $\subseteq [5'10'', 6']$). Moreover, a dynamic version of interval semantics is needed to analyze sentences like *Mary is taller than exactly three boys are* (Zhang 2020).

less likely than alternatives. Rather than addressing who spit the thistles out or who was the least likely to do so, (91) addresses a degree QUD: *how prickly the thistles are*. Compared to its alternatives (e.g., 'Pooh spit them out'), the prejacent 'Eeyore spit them out' is maximally informative in resolving this degree QUD.

(91) (Context: Imagine Pooh and friends coming upon a bush of thistles. Eeyore (known to favor thistles) takes a bite but spits it out. (see Szabolcsi 2017))
Those thistles must be really prickly! Even Eeyore$_F$ spit them out!

(92) The presuppositions of *even*(p) (from Zhang 2022; see also Greenberg 2018):
a. there is a contextually salient degree QUD
b. p is maximally informative in resolving the degree QUD

Thus the current analysis predicts that *even* can be used in a cumulative-reading sentence or a multi-head comparative to address the same underlying degree QUD. This prediction is borne out. As illustrated in (93), both the use of *even* and the use of numerals or changes address a QUD such as the skewness of wealth distribution:[18]

(93) a. At most 3% of the population owns even more than 70% of the land.
b. Fewer and fewer people own even more and more wealth.

6 Conclusion

Based on our understanding of well-known phenomena – comparatives and cumulative-reading sentences, this paper has analyzed the semantics and pragmatics of multi-head comparatives. These constructions express changes along multiple dimensions, and the interplay between those changes addresses a single underlying issue. I have developed the notion of 'degree-QUD-based informativeness' to capture this interplay, showing how multi-dimensional information is integrated to address interlocutors' conversational goals.

Multi-head comparatives shed light on how human cognition manages complex mathematical operations that involve multi-dimensional information. Thus, the current work also echoes existing research on intuitive mathematical computations encoded in language (e.g., Zhang and Ling 2021 on interval subtraction, Coppock 2022a on division, Coppock 2022b on percentage).

[18]I thank Sigrid Beck for pointing out this to me.

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